IMPLEMENTING TOURISM CIRCUIT CONCEPT WITH MACHINE LEARNING ON A WEB-BASED TOURISM PLATFORM FOR MAJAYJAY, LAGUNA

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The Researchers

DEDICATION

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ABSTRACT

Tourism plays a vital role in local economic development, especially in areas with rich cultural and natural attractions. Majayjay, Laguna, known as a hidden gem, continues to face challenges in promoting its tourist spots because of limited online presence and fragmented information. To address this, a web-based tourism platform with machine learning was developed to improve the visibility, accessibility, and usability of the Majayjay tourism circuit. The system features user registration, accommodation listings, booking, and personalized recommendations that guide tourists in planning their visits. It was built using React.js for the frontend, Node.js for backend operations, Flask (Python) for the machine learning component, Leaflet.js for interactive mapping, and MySQL for data management. The machine learning model was trained and deployed locally, ensuring flexibility and independence from third-party platforms. By integrating interactive maps and intelligent recommendations, the platform enhances the tourist experience while promoting sustainable tourism practices. Ultimately, this project demonstrates how digital innovation can strengthen community-based tourism and create opportunities for local economic growth. Results showed that the system functioned correctly, reliably, and efficiently. Compatibility testing across devices and browsers showed consistent performance, while user feedback indicated satisfaction with usability, usefulness, security, and portability, particularly appreciating the interactive map and personalized recommendations. Overall, the Majayjay Tourism Circuit platform is technically reliable, user-friendly, and effective in promoting tourism. By integrating intelligent recommendations and interactive mapping within a flexible, user-centered design, it enhances tourist experiences, supports community-based tourism, and contributes to local economic growth, successfully positioning Majayjay as a hidden gem for wider audiences.

Keywords: *Tourism Circuit, Web-Based Platform, Machine Learning, Sustainable Tourism, Majayjay Laguna, Interactive Mapping, Community-Based Tourism*

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Definition of Terms

Accommodation Module	A system component that manages lodging listings in Majayjay, Laguna, enabling users to browse and book available accommodations.
Admin Module	A system feature that provides administrators with access to verify listings, manage user accounts, and monitor system activities.
Database.	A structured electronic repository where the system stores and retrieves information such as user profiles, tourist spots, accommodations, and booking records.
Geographic Information System (GIS) Mapping	A digital mapping feature that displays the location of tourist attractions, accommodations, and landmarks in Majayjay.
Machine Learning (ML)	A field of artificial intelligence (AI) that enables the system to identify patterns from data and generate personalized recommendations for users.
Recommendation System	A feature of the platform that applies machine learning algorithms to suggest tourist destinations, accommodations, and activities based on user preferences.
Sustainable Tourism	A tourism practice that minimizes environmental impact, supports local communities, and ensures long-term economic and cultural benefits.
Tourism Circuit	An interconnected route of tourist attractions, accommodations, and services within Majayjay, Laguna, designed to create a continuous travel experience.
User Interface (UI)	The visual and interactive layer of the system where users access and navigate the platform's features.
Web-Based Platform	An online system accessible through a web browser without installation, designed to deliver tourism-related services to users.

CHAPTER 1

INTRODUCTION

Project Context

Tourism is one of the largest and fastest-growing industries in the world, contributing significantly to employment, livelihood, and national income. According to the Philippine Statistics Authority (PSA), the country's tourism industry accounted for 8.6% of the national GDP in 2023, highlighting its importance in economic development. During the same year, the Philippines welcomed over 5.45 million international visitors, along with millions of domestic travelers. These numbers reflect how tourism directly supports livelihoods and contributes to national income, especially in provinces with rich cultural and natural resources.

To further strengthen the local tourism economy and create structured travel experiences, many regions adopt what's called a tourism circuit. A tourism circuit is a route that connects multiple tourist destinations, allowing travelers to experience a variety of attractions in a structured and efficient manner.

Typically, a tourism circuit comprises at least three major destinations located in different towns or cities, situated close enough to each other to be visited sequentially. These circuits are designed with well-defined entry and exit points to facilitate smooth travel and enhance the overall tourist experience (Tourism Beast, n.d.).

One of these promising locations is Majayjay, Laguna, a small, peaceful town known for its cool weather, natural springs, waterfalls like Taytay Falls and Bukal Falls, and historical landmarks like the centuries-old Saint Gregory the Great Parish Church (Majayjay, 2022). Despite its beauty and charm, Majayjay is still considered a place that many people have yet to discover. With its rich natural resources and culture, it is perfect for a sustainable and eco-friendly tourism circuit (Karol, 2016).

Ecotourism is a kind of tourism that focuses on visiting natural places while helping to protect the environment and support local communities. It encourages tourists to travel responsibly, respect nature, and learn about local traditions (GSTCouncil, 2022). When done properly, ecotourism and sustainable tourism can create jobs, support local businesses, and preserve the beauty of places like Majayjay for future generations. This supports global goals like the United Nations' Sustainable Development Goals (SDGs), especially SDG 8: Decent Work and Economic Growth.

As tourism grows, technology is becoming a big part of how people travel. From searching places online to booking hotels through apps, tourists now depend on technology to plan their trips. This is where Information Technology (IT) becomes important. By turning the tourism circuit of Majayjay into a digital platform, we can make it easier for people to discover and explore the town. A web-based tourism platform can help tourists find destinations, book accommodations, get travel tips, and even receive personalized recommendations based on their interests.

This platform can also use GIS mapping to show exact locations of tourist spots on a map, so travelers won't get lost. It can include machine learning (ML) features that suggest places based on what the user likes (Singh, 2019). Locals can also list their homes or eco-lodges, like how Airbnb works, so they can earn income while offering authentic experiences. The system may include online payments, tourist reviews, and eco-friendly badges that promote responsible travel choices.

By combining tourism and technology, we can help a hidden gem like Majayjay reach more people while protecting its environment and culture. This not only supports tourism but also helps the community grow in a sustainable way. Turning the tourism circuit into a digital platform is a smart solution that connects nature, people, and technology—all working together for a better future.

Project Objectives

The general objective of this capstone project is to implement the Tourism Circuit concept using Machine Learning within a web-based tourism platform for Majayjay, Laguna. This initiative aims to promote sustainable tourism, improve visitor experience, and support local communities through intelligent digital solutions.

Specifically, the project seeks to:

1. To develop a functional and reliable web platform with core modules for user accounts, tourism listings, lodging information, and an administrative dashboard.

2. To implement a smart recommendation engine using machine learning to provide personalized suggestions for tourist spots and activities.

3. To evaluate the platform's overall quality and user acceptance by applying both the ISO 25010 standard for technical quality and the Technology Acceptance Model (TAM) for perceived usefulness and ease of use.

Project Purpose

The primary purpose of this project is to enhance tourism in Majayjay, Laguna by leveraging digital technology to promote sustainable tourism. This platform would empower local businesses, improve access to tourism-related services, and provide new economic opportunities for the community. By focusing on sustainable practices, it ensures minimal environmental impact while contributing to the local economy.

This project directly aligns with Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth, which aims to promote inclusive economic growth, provide productive employment, and foster decent work opportunities for all.

Through this platform, the local community, specifically small business owners, local tour guides, and residents, would benefit by having better visibility, more business opportunities, and an increased flow of tourism. Moreover, it would help achieve economic sustainability by encouraging responsible tourism and supporting locally-owned establishments. The beneficiaries of this project include:

Local Entrepreneurs. Small business owners in Majayjay, such as homestay operators, tour guides, and local vendors, would gain increased visibility to a broader audience, both local and international tourists. This exposure would allow them to connect directly with travelers, increasing the likelihood of bookings and sales. As a result, these businesses would benefit from more stable income and greater financial security, fostering local economic growth.

Local Government. The local government would have better oversight of tourism activities, ensuring that tourism development is sustainable and beneficial to the community. With improved access to data and trends, they can make informed decisions to prevent overburdening local resources, ensuring that tourism aligns with sustainable practices while promoting the overall well-being of the community.

Tourist. Both local and foreign visitors would find it easier to plan and enjoy their trips. By having access to a wealth of information about trusted accommodations, tour guides, and eco-friendly activities, tourists would be able to make informed choices tailored to their interests and needs. This would result in a smoother and more efficient travel experience, enhancing their overall satisfaction.

Future Researchers. This project can serve as a foundation for future studies on sustainable tourism, digital platforms, and their impacts on local economies. The system's design, which integrates GIS Mapping, Recommendation Systems, and Sustainability Features, provides a practical case for researchers to explore how technology can assist small towns like

Majayjay in growing economically without compromising their cultural and environmental heritage. Researchers can also explore how machine learning and digital platforms can shape the future of tourism.

Scope and Limitation of the Study. The system aims to promote the town's tourism attractions, support local businesses, and provide tourists with a smart and user-friendly way to plan their visit. The platform would help tourists discover Majayjay as a hidden gem while also supporting Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth by creating more opportunities for local tourism-related jobs and services.

The system would include the following features and modules:

Account Management Module. Allows tourists and administrators to create and manage their accounts. New users can easily sign up by providing basic information such as their name, email address, and password, while returning users can log in securely to access their profiles. After logging in, users can log out anytime to protect their account security.

Tourists, once logged in, can explore tourist destinations, view maps with locations pinned, receive personalized recommendations based on their preferences, and leave reviews or ratings for tourist spots and stays.

Administrators, on the other hand, have the highest level of access, allowing them to update information about tourist spots, and ensure that the platform is running smoothly, accurately, and securely.

Tourist Listing Module. Provides tourists with a comprehensive guide to explore the cultural, eco-friendly, and tourism aspects of Majayjay, Laguna. Users can view detailed information about various tourist spots, including their names, descriptions, eco-friendly features, and recommended visiting hours. Through an interactive GIS map, tourists can see the exact locations of attractions, accommodations, and routes, making it easy to plan their trips.

Additionally, this module highlights local events, festivals, and traditions by providing a cultural calendar, helping tourists immerse themselves in the community's heritage and participate in activities that align with their travel interests. The module also promotes responsible tourism by offering eco-friendly travel tips and encouraging users to leave reviews and ratings for the places they visit. This feedback not only helps future tourists but also supports sustainable tourism practices by emphasizing the importance of preserving cultural and natural resources.

Accommodation Management Module. Allows tourists to browse available accommodations such as homestays, inns, and bed and breakfasts within Majayjay. Tourists can view important details like pricing, contact information, and pictures to help them choose where to stay.

Administrators have the ability to manage and update accommodation listings, ensuring that all information remains accurate and up-to-date. This module ensures that tourists can easily find places to stay without direct involvement from accommodation providers.

Recommendation Module. Designed to make the platform more personalized and user-friendly for each tourist. It analyzes a tourist's past activities, preferences, and interactions with the system to offer tailored suggestions for tourist spots and activities, It also takes in account for the current weather and season. For example, if a tourist has shown interest in nature or eco-friendly attractions, the module would recommend similar locations and activities that align with their interests. This not only makes it easier for tourists to find places and activities they would enjoy, but it also helps them discover new spots they may not have thought of on their own.

For tourists who are new to Majayjay, Laguna, or unsure about where to start, this module simplifies the decision-making process by providing a generalized suggestion that can guide them through their trip planning by showing them the best places to visit in the current season. To making the experience more intuitive, the Recommendation Module ensures that every tourist feels like the platform understands their preferences and can help them create a memorable and enjoyable visit.

Admin Dashboard Module. The Admin Dashboard is designed for platform administrators to manage overall operations. Only administrators have the authority to create and update accommodation listings, ensuring all details are accurate and verified. In addition, admins can manage tourist spot information, monitor user accounts, update eco-travel tips, and maintain platform content. This central control guarantees that the platform remains reliable, organized, and high-quality for all users.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature

This chapter reviews existing literature on digital tourism platforms, sustainable tourism, the tourism circuit and the role of machine learning in the tourism circuit. It provides the foundational knowledge that supports the rationale, development, and implementation of the proposed Web-Based Sustainable Tourism Platform for Majayjay, Laguna.

Sustainable Tourism in the Philippines

Ecotourism has emerged as a pivotal mechanism for fostering environmental, socio-economic, and cultural values at both local and national levels, particularly in the context of the Pandin Lake Tourist Service Cooperative in San Pablo City, Laguna. According to Patrick James and G. Putungan (2023), the cooperative serves as a social enterprise that not only enhances the local economy through tourism but also aligns with the principles of Gender and Development (GAD) by collecting Sex Disaggregated Data (SDD) and Gender Disaggregated Data (GDD). This data is crucial for understanding the demographic profile of cooperative members and for informing government programs aimed at sustainable tourism development. The study emphasizes ecotourism as a sustainable source of income that creates jobs and supports community development while highlighting the need for effective cooperative management. It also underscores the importance of government policies, such as the Tourism Act of 2009, in promoting sustainable tourism and environmental conservation.

Bersales et al. (2019) emphasize the importance of sustainable tourism indicators in assessing the environmental impacts of tourism in the Philippines. While tourism supports economic growth, the study highlights its potential for environmental damage and proposes a methodology using the Philippine Input-Output tables and the Philippine Tourism Satellite Accounts (PTSA) to measure indicators such as energy use, water consumption, carbon emissions, and waste generation. The study underscores the need for a consistent accounting framework to support policy evaluation and guide sustainable tourism planning aligned with environmental stewardship..

Calinao et al. (2021) highlight the importance of community participation in promoting environmental sustainability amid challenges brought by modernization and globalization. Their descriptive study in Calamba, Laguna shows that involving local communities increases environmental awareness, improves relationships between residents and tourists, and enhances overall quality of life. The study identifies spontaneous, induced, and coercive forms of participation, linking them to economic, social, and environmental benefits. It further emphasizes the role of LGUs in implementing programs that support sustainability, job creation, and social harmony through active community engagement. The profitability and sustainability of farm tourism businesses in the Philippines, particularly in Angeles City and Bacolor, Pampanga, have been critically examined by De Leon et al. (2024). This study highlights the significant role of farm tourism in contributing to the local economy, especially in the aftermath of the COVID-19 pandemic, which severely impacted traditional agricultural practices. The research reveals that while farm tourism businesses were profitable, with an average profit of Php 2,388,500.00 in 2019, this figure

declined to Php 1,224,750.00 in 2023 due to the lingering effects of the pandemic.

The study employs the theory of the firm alongside the concepts of economic, social, and environmental sustainability to assess farm tourism practices. It identifies socially sustainable practices such as hiring local workers, providing flexible work schedules, and offering free housing and food, which contribute to community well-being. Environmentally, the farms implement practices like organic farming, composting, and maintaining a carrying capacity of 69 visitors per hectare, which align with sustainable tourism principles. The findings underscore the importance of farm tourism not only as a source of income but also as a means of promoting sustainable agricultural practices and community development. The study advocates for expanded financing options for farm improvements and innovation, emphasizing the need for a holistic approach to enhance the sustainability of farm tourism in the region.

The development of sustainable tourism, particularly ecotourism, in rural areas like Rizal, Laguna, has been examined by Sarinas et al. (2023). The study highlights the potential of ecotourism to stimulate socio-economic development and foster environmental conservation. However, it also identifies various issues and challenges associated with ecotourism, such as negative environmental impacts, depletion of forests, disturbance of wildlife, and “greenwashing.” The research employs a gap analysis to evaluate the current state of TANAW de Rizal Park in relation to the criteria for sustainable ecotourism established in the National Ecotourism Strategy and Action Plan

2013-2022. The findings reveal that while the park generally aligns with the attributes of an ecotourism destination, there are areas for improvement, particularly in terms of community involvement, capacity building, and environmental education. The study emphasizes the importance of active community participation, skills development, and environmental education to ensure the sustainability of ecotourism development. The results of this study provide direction to the local government unit (LGU) in developing the area into a sustainable ecotourism destination.

The relevance of sustainable touri”m in’the Philippines, particularly in the context of Boracay Island and Puerto Princesa City, has been explored by Manalo (2017). The study emphasizes the dual role of tourism as a significant contributor to economic development while also posing challenges to environmental sustainability. It highlights the shift towards sustainable tourism practices as a necessary response to the negative impacts of traditional tourism, such as environmental degradation and community disruption. The research underscores the importance of local government units (LGUs) in implementing policies that promote sustainable tourism, as evidenced by the establishment of the Puerto Princesa Subterranean River National Park as one of the New Seven Wonders of Nature. The findings indicate that both Boracay and Puerto Princesa have made strides in integrating sustainable practices, such as community-based participation and the use of green technologies, to enhance their ecotourism offerings. However, the study also identifies gaps in the current practices, particularly in community involvement and the need for effective governance to ensure equitable benefits from tourism. The insights provided by this research are crucial for guiding future sustainable tourism

initiatives in the Philippines, emphasizing the need for collaboration among stakeholders to achieve long-term sustainability goals.

The strategic marketing framework for community-based ecotourism (CBET) in Panguil, Laguna, as presented by Ruiz et al. (2019), underscores the importance of sustainable tourism practices in promoting local economic development while preserving natural resources. The study highlights the potential of the Panguil River Ecopark (PREP) as a model for CBET, emphasizing the need for a comprehensive approach to mitigate environmental degradation. Utilizing a combination of quantitative and qualitative methods, including surveys, focus group discussions, and participatory action research, the authors identify critical gaps in the current ecotourism practices at PREP.

Key findings indicate that while the park possesses rich cultural and natural resources, issues such as inadequate marketing capacity, lack of community involvement, and failure to implement carrying capacity measures pose significant challenges to sustainable development. The study advocates for the introduction of diverse tourism products, community engagement in decision-making, and the establishment of a strategic marketing framework to enhance the sustainability of ecotourism in the region. By addressing these gaps, the research aims to provide actionable recommendations for local government units (LGUs) to develop Panguil as a sustainable ecotourism destination, thereby contributing to the broader goals of sustainable tourism in the Philippines.

Mandigma Jr., Ruiz, and De Vero (2019) explores the destination competitiveness of beach tourism in Batangas Province, Philippines, with a

focus on sustainability. The research highlights the critical role of tourism in economic development while addressing the environmental and socio-cultural challenges associated with tourism growth. Utilizing a quantitative approach, the study assesses the sustainability of beach destinations in Batangas in terms of economic viability, socio-cultural impact, and environmental preservation. Findings indicate that while the beach destinations are perceived as competitive, there are significant gaps in areas such as marketing capacity, community involvement, and adherence to carrying capacity guidelines. The study emphasizes the importance of developing a sustainable tourism development plan that incorporates diverse tourism products, community engagement, and effective marketing strategies. By aligning with the principles of sustainable tourism, the research aims to provide actionable recommendations for local government units (LGUs) to enhance the competitiveness of beach tourism in Batangas, ultimately contributing to the sustainable development of the region.

Sandhu and Berse (2022) introduces a System of Systems (SoS) approach to address the complex interrelated challenges faced by coastal cities in the Philippines, particularly in the context of sustainable tourism and disaster risk management. The authors emphasize the dual role of tourism as a significant economic driver while also highlighting its vulnerability to natural disasters and environmental degradation. The paper argues for the necessity of integrating sustainable development goals (SDGs) into tourism governance, advocating for a holistic understanding of the interconnectedness of economic, social, and environmental factors. By applying systems theory, the study illustrates how coastal city managers can enhance governance through

improved evidence gathering and feedback mechanisms, ultimately leading to more effective decision-making. The findings underscore the importance of collaboration among local government units (LGUs), stakeholders, and communities to build resilience against disasters while promoting sustainable tourism practices. This research contributes to the growing discourse on sustainable tourism management in the Philippines, providing a framework for addressing the multifaceted challenges of coastal urban environments.

Manalo (2017), the transformative journey of Puerto Princesa City towards becoming a model of sustainable tourism is examined, highlighting the critical role of effective policy implementation in achieving economic development while preserving natural resources. The study emphasizes the significance of sustainable tourism as a major source of income for local governments, particularly through the management of the Puerto Princesa Subterranean River National Park (PPSRNP), which gained international recognition as one of the New Seven Wonders of Nature. The research outlines the impact of City Ordinance No. 163-91, which prohibits littering and promotes cleanliness, as a foundational policy that has led to improved waste management practices and enhanced the city's image as a tourist destination.

The findings indicate that the strong political would of local government officials, combined with community participation, has been instrumental in fostering environmental awareness and conservation efforts among residents. This case study serves as a valuable example for other local government units (LGUs) in the Philippines, demonstrating how sustainable tourism practices can lead to significant socio-economic benefits while ensuring the protection of the

environment. The paper concludes with recommendations for further enhancing sustainable tourism initiatives in Puerto Princesa, emphasizing the need for continued community engagement and effective policy frameworks.

Borbon (2020) conducts a needs assessment for a municipality in the Philippines to propose a sustainable community extension tourism program aimed at enhancing local development and improving the quality of life for residents. The research employs a descriptive method, gathering data from local residents to identify their needs across four key areas: economic, health, socio-cultural, and environmental. Findings reveal that the community has a significant need for environmental initiatives, followed by health and economic support, while socio-cultural needs are moderately prioritized. The study emphasizes the importance of community engagement in developing sustainable tourism initiatives, highlighting that effective policy implementation can lead to improved livelihoods and environmental stewardship. By proposing a structured community extension program that includes youth empowerment, health care awareness, and environmental protection activities, the research underscores the potential for tourism to serve as a catalyst for sustainable development in rural areas. This work contributes to the broader discourse on sustainable tourism by demonstrating how targeted community programs can address local needs while fostering economic growth and environmental conservation.

Shih (2013) explores the implications of collaborative network governance for sustainable tourism development in Bulacan Province, Philippines. The research highlights the rapid growth of the tourism industry globally and the

associated challenges, particularly in developing countries, where unplanned tourism development can lead to socio-cultural and environmental degradation. Shih argues that collaborative governance can address the fragmented nature of tourism development by fostering cooperation among various stakeholders, including local government units (LGUs), private sector actors, and community organizations. The study identifies key challenges in Bulacan, such as political influence and inadequate funding, which hinder effective tourism governance. It emphasizes the need for a coordinated approach to tourism planning that integrates sustainability principles and encourages stakeholder participation. The findings suggest that while a fully collaborative network governance model may not be feasible in the current political climate, informal cooperative and coordinative governance structures could be more effective in promoting sustainable tourism initiatives. This research contributes to the understanding of governance dynamics in tourism planning and underscores the importance of local context in shaping sustainable tourism outcomes.

Tourism Circuit

The concept of a tourism circuit is essential in organizing and promoting tourism in a sustainable and inclusive way. A tourism circuit refers to a planned and connected route that links multiple tourist destinations which are often near each other or share a common theme. These circuits are designed to provide travelers with a richer and more diverse experience, encouraging them to explore several sites rather than visiting just one location. This idea helps maximize the tourism potential of an entire area instead of focusing only on a popular spot.

According to Shukla, A (2024) and Kumar & Goel (2017), tourism circuits are crucial for regional development because they spread the benefits of tourism to smaller communities and lesser-known destinations. Tourism circuits can be categorized into different types depending on their purpose and design. Thematic circuits are based on specific interests such as ecotourism, religious tourism, wellness, or culinary experiences. These circuits are attractive to tourists who have focused preferences and want to explore destinations tied to a certain theme. For instance, an eco-tourism circuit may include waterfalls, hiking trails, and nature parks, while a heritage circuit may focus on ancestral houses, historic churches, and cultural landmarks. Geographical circuits, on the other hand, are based on physical location and convenience, connecting sites that are close to each other to reduce travel time. Cultural circuits link areas with shared traditions, festivals, or historical significance (Sharma, 2018; Chakrabarty & Dey, 2020).

In recent years, the idea of smart tourism circuits has emerged. This refers to using digital technology and data to enhance the tourist journey. These circuits integrate apps, smart maps, virtual tours, and real-time updates to improve visitor experience and decision-making. With the help of technology, tourists can easily find directions, receive personalized recommendations, and learn more about the places they visit. As Buhalis & Amaranggana (2015) explain, this shift toward smart tourism supports not just the travelers but also helps destinations manage tourist traffic better and make data-driven improvements. This is especially valuable in remote and developing areas that may not have traditional tourism infrastructure.

The value of tourism circuits is particularly high in areas like Majayjay, Laguna, which is rich in natural beauty and cultural heritage but remains unexplored. Known for its waterfalls, centuries-old church, mountain views, and rural charm, Majayjay has the potential to attract more tourists through a well-planned tourism circuit.

Currently, most visitors only go to one or two famous spots like Taytay Falls. But by creating a circuit that includes nearby trails, river spots, local farms, artisan markets, and accommodations, tourists can be encouraged to stay longer and spend more—benefiting the whole community. This strategy helps promote Majayjay not just as a single destination but as part of a broader tourism experience.

Tourism circuits also align with sustainable tourism goals, especially in protecting the environment, promoting local culture, and supporting community livelihoods. When tourist visits are distributed across several sites, it prevents overcrowding and reduces environmental damage to a single location. At the same time, more communities gain access to tourism-related jobs and income. This inclusive tourism model supports the UN Sustainable Development Goals (SDGs), especially SDG 8 (Decent Work and Economic Growth), SDG 11 (Sustainable Cities and Communities), and SDG 12 (Responsible Consumption and Production). According to the United Nations World Tourism Organization (UNWTO, 2016), circuits are key to making tourism more equitable, especially in developing countries where rural tourism can significantly impact livelihoods.

In the Philippines, the Department of Tourism (DOT) has recognized the value of tourism circuits in its National Tourism Development Plan. Initiatives

like the “Tourism Circuit Development Project” support local government units in identifying attractions, creating infrastructure plans, and organizing routes that link tourist sites with accommodations, restaurants, and transportation. In regions like Laguna, this approach is essential in highlighting less-commercialized municipalities like Majayjay. The goal is to develop tourism that reflects the local identity and preserves cultural heritage while improving access and amenities for visitors (DOT, 2021).

Community-based tourism, which involves locals in decision-making and operation, is also part of this circuit model and ensures that tourism development is inclusive and empowering.

The integration of Information Technology (IT) into tourism circuits brings additional advantages. Using platforms with built-in Geographic Information Systems (GIS), tourism managers can create interactive maps, track tourist movements, and analyze visitation patterns. Machine Learning, a part of Artificial Intelligence, can be used to predict tourist preferences and recommend sites or activities based on previous behavior or user interests. This creates a more customized experience for tourists while helping authorities manage destinations more efficiently. For example, if a particular waterfall gets overcrowded during weekends, the system can recommend nearby alternative sites to reduce pressure on the environment. This kind of smart management makes tourism more enjoyable and more sustainable at the same time (Neuhofer et al., 2015; Ranjan, 2019).

Moreover, digital tourism circuits make travel more accessible to tech-savvy tourists who use smartphones and apps for planning. A web-based

platform like the one proposed in this study can serve as a digital gateway to Majayjay's tourism circuit. It can showcase multimedia content, highlight suggested routes, and offer integrated bookings for accommodations or tour guides. With added features like peer reviews, payment integration, and personalized recommendations using machine learning, the system can guide tourists efficiently while empowering local entrepreneurs and service providers.

The Role of Machine Learning in Tourism Circuit

Machine learning (ML) plays a transformative role in modern tourism, especially in optimizing tourism circuits. A tourism circuit is a connected route that links multiple attractions, often within a defined geographical area, to promote regional development and enhance the visitor experience. With the advancement of ML technologies, tourism circuits can be intelligently managed, promoted, and enhanced through data-driven insights and predictive analytics.

One of the most common applications of ML in tourism circuits is personalized recommendations. ML algorithms can analyze user behavior, preferences, past bookings, and feedback to suggest destinations, accommodations, and activities suited to individual travelers (Gretzel et al., 2015; Gretzel, U., `Zhong, et al., 2016). For instance, if a tourist previously favored eco-tourism spots or cultural heritage sites, the system can tailor suggestions accordingly using collaborative filtering or content-based filtering algorithms.

Machine learning techniques have proven to be invaluable for forecasting tourist behavior and visitor demand. For example, a study by Mishra et al., (2021) explored machine learning-based forecasting systems to predict tourism

patterns, enabling better resource allocation and enhancing overall tourism management. This helps in planning itineraries that avoid overcrowding and promote lesser-known spots, which is vital for sustainable tourism circuit development.

ML also enhances natural language processing (NLP) capabilities within tourism platforms. Chatbots, powered by NLP and ML, provide 24/7 assistance in planning, booking, and answering tourist queries, improving user experience and operational efficiency (Sigala, 2018). These bots can even communicate in multiple languages, making the platform accessible to international visitors.

In the context of image recognition, ML helps identify and tag images of tourist destinations uploaded by users, which in turn boosts content engagement and automates the curation of travel galleries (Khan et al., 2020). Moreover, ML-powered sentiment analysis of online reviews and social media posts can be used to gauge tourist satisfaction and identify issues that require attention.

For geospatial data analysis, ML can integrate with Geographic Information Systems (GIS) to analyze location-based patterns and optimize route planning in the tourism circuit. Tools like K-means clustering and decision trees can segment tourist types based on travel behavior and suggest optimal paths across attractions in Majayjay, Laguna (Baggio et al., 2020).

Furthermore, ML aids in market segmentation by identifying groups of tourists with similar behaviors or preferences. This segmentation enables targeted marketing and development of tailored experiences for specific

demographics, such as eco-tourists, cultural travelers, or adventure seekers (Xu et al., 2021).

Machine learning also supports dynamic pricing models, where accommodation and travel costs can be adjusted based on demand prediction, seasonality, and user behavior. This benefits both local businesses and tourists by balancing affordability and profitability.

Finally, the evaluation of tourism initiatives becomes more effective with ML-driven analytics. By aggregating data from multiple sources (e.g., user feedback, transaction logs, GPS tracking), ML can help evaluate which parts of a tourism circuit are underperforming or succeeding, aiding tourism officers and local government units (LGUs) in decision-making.

Forecasting tourist arrivals has become increasingly important for economic planning and policy-making in the tourism sector, as it significantly contributes to GDP and employment worldwide (Gunter & Onder, 2015). Traditional forecasting methods, such as Autoregressive Integrated Moving Average (ARIMA) models, have been widely used; however, they often fall short compared to advanced artificial intelligence techniques, which can capture complex, non-linear relationships in the data (Song & Li, 2008). Recent studies have demonstrated that incorporating internet search data, such as Google Trends and Baidu Index, into forecasting models can substantially enhance accuracy by reflecting real-time consumer interest and behavior (Yang et al., 2015). The introduction of Kernel Extreme Learning Machines (KELM) represents a significant advancement in this field, as these models leverage

both traditional time series data and search engine indexes to improve forecasting performance.

Personalized travel recommendation systems (TRS) leverage machine learning algorithms to enhance the tourism experience by analyzing user data to provide tailored suggestions for destinations, attractions, accommodations, and activities based on individual preferences and behaviors (Badouch & Boutaounte, 2023). These systems utilize various approaches, including collaborative filtering, which identifies similarities between users based on past interactions, and content-based filtering, which recommends items based on their attributes and user preferences. Hybrid systems that combine both methods have proven effective in generating accurate recommendations. The benefits of TRS include improved personalization, convenience, and user experience, ultimately leading to increased revenue for tourism businesses. By continuously learning from new data and user feedback, these systems can adapt and refine their recommendations, making them invaluable tools for travelers seeking personalized and relevant travel options (Badouch & Boutaounte, 2023).

The phenomenon of overtourism poses significant challenges for tourist destinations, particularly in Spain, where the rapid growth of tourism has led to overcrowding and associated negative impacts on local communities (Perles-Ribes et al., 2020). This study explores the application of machine learning techniques as a predictive tool for identifying potential over tourism situations in various Spanish cities, utilizing a dataset that includes indicators of tourism competitiveness, pressure from traditional and sharing economy

accommodations, and economic dependence on tourism. The research highlights the importance of developing an early warning system to manage tourism flows effectively, as traditional qualitative approaches have often failed to provide actionable insights. By employing advanced machine learning methods, the authors aim to establish a more robust framework for predicting overtourism, thereby enabling destination managers to implement strategies that balance tourism growth with the well-being of local residents (Perles-Ribes et al., 2020). The findings suggest that increased tourism pressure, particularly from the sharing economy, correlates with higher risks of overtourism, underscoring the need for comprehensive management strategies that consider both economic benefits and social impacts (Perles-Ribes et al., 2020).

The application of machine learning techniques in predicting visitor numbers to the Moche Route in Peru represents a significant advancement in tourism management, particularly in the context of post-COVID-19 recovery (Bravo et al., 2023). This study employs a structured methodology encompassing data collection, model analysis, development, and evaluation, utilizing public data from official sources and online platforms like TripAdvisor and Google Trends from January 2011 to May 2022. The research evaluates four machine learning algorithms—linear regression, KNN regression, decision tree, and random forest—finding that linear regression yields the most accurate predictions for both national and foreign tourists.

The results underscore the potential of machine learning to facilitate digital transformation in the tourism sector, enabling better management strategies that can enhance visitor experiences while promoting sustainable

tourism practices (Bravo et al., 2023). By leveraging diverse data sources and advanced predictive models, this research contributes valuable insights for tourism authorities aiming to revitalize the Moche Route and improve local economies.

The study conducted by Li (2022) explores the application of machine learning techniques to predict visitor numbers to the Moche Route in Peru, a region rich in cultural heritage and archaeological significance. In light of the challenges posed by the COVID-19 pandemic, the research emphasizes the need for effective management strategies to revitalize the tourism sector through digital transformation and the use of emerging technologies. By employing a methodology that includes data collection, model analysis, development, and evaluation, the study utilizes historical visitor data alongside online sources such as TripAdvisor and Google Trends to enhance prediction accuracy. The research evaluates several machine learning algorithms, ultimately finding that the Sparse Principal Component Analysis-Convolutional Neural Network-Long Short-Term Memory (SPCA-CNNLSTM) model outperforms traditional methods, providing valuable insights for tourism managers to optimize visitor experiences and resource allocation (Li, 2022). This innovative approach not only contributes to the understanding of visitor patterns but also supports sustainable tourism development in the region.

The research conducted by Rubi et al. (2022) investigates consumer behavior regarding the purchase of travel insurance, utilizing machine learning techniques to predict whether individuals are likely to buy insurance based on various demographic factors.

The study analyzes a dataset of 3,980 travelers, focusing on independent variables such as age, employment type, annual income, family size, and health conditions. The authors apply ten different classification algorithms, including Random Forest, Decision Tree, and Stochastic Gradient Descent, to determine the most accurate model for predicting travel insurance purchases. The findings reveal that Random Forest, Decision Tree, and Stochastic Gradient Descent achieve the highest accuracy rates of 88%, indicating their effectiveness in identifying potential insurance buyers. This research highlights the importance of leveraging machine learning in the insurance industry to enhance targeted marketing strategies and optimize resource allocation, ultimately benefiting both consumers and insurance providers (Rubi et al., 2022).

Conversely, Núñez, Gómez-Pulido & Ramírez (2024) sheds light on the challenges faced when integrating ML into the tourism sector. The review emphasizes issues such as the heterogeneity of data sources, the dynamic nature of tourism demand, and the complexity of delivering personalized experiences at scale. These challenges can hinder the effectiveness of ML models, especially when data quality and integration are suboptimal. The study suggests that while ML holds promise, its application requires careful consideration of these inherent complexities within the tourism industry.

Synthesis and Relevance of the Study

In synthesizing the various theories and related literature reviewed, this study draws upon a convergence of concepts from sustainable tourism, tourism circuit theory, and machine learning applications. Sustainable Tourism emphasizes environmental, cultural, and economic preservation, principles

which are directly aligned with the goals of promoting Majayjay, Laguna as a sustainable destination.

The Tourism Circuit theory complements this by advocating for interconnected points of interest, thereby supporting the creation of a structured yet dynamic travel experience within Majayjay.

Machine Learning, as another central pillar of this research, is leveraged to enhance the personalization of travel experiences. With algorithmic analysis of user preferences, tourism trends, and geographic data, the system is designed to generate intelligent travel recommendations that respond to user needs and interests.

The integration of Geographic Information Systems (GIS), user-generated reviews, and local tourism data within a machine learning framework marks a significant innovation in tourism management for small communities. By localizing the platform to Majayjay, the system not only highlights hidden gems such as Bukal Falls and Taytay Falls but also addresses the digital divide by offering a smart and accessible solution that bridges traditional tourism with modern digital innovation.

This research is highly relevant in today's context where tourism is slowly recovering post-pandemic, and where local tourism circuits need technological support to reach wider audiences. It also addresses the lack of digital infrastructure for promoting rural destinations like Majayjay, enabling local businesses, accommodations, and guides to be part of an inclusive tourism economy.

The study contributes to both academic discourse and practical applications. Academically, it validates the application of machine learning in a localized tourism context. Practically, it offers a working prototype that may be adopted by local government units (LGUs) and tourism offices to enhance tourism experiences, support sustainable practices, and increase tourist arrivals in Majayjay, Laguna.

Literature Matrix

Table 1. Summary of Literature

Author/s & Year	Title	Methodology	Findings
Patrick James & G. Putungan (2023)	Ecotourism, Social Enterprise of the Pandin Lake Tourist Service Cooperative, Sto. Angel. San Pablo City, Laguna Assessment and Tourism Development Plan	Assessment and Development Plan	Ecotourism projects like Pandin Lake are viable and sustainable if properly managed with strong community participation and gender sensitivity, contributing both to livelihood and environmental preservation.
Bersales, Ilarina, Porquiz, Caranay (2019)	Towards the Development of Sustainable Tourism Indicators for the Philippines	Provisional methodology for indicators	A framework of sustainable tourism indicators must be standardized across regions to ensure balanced economic growth and minimize environmental degradation.
Calinao, Caparas Celino, Nualla (2021)	Community Participation and Environmental Sustainability: A Study in Calamba, Laguna	Descriptive study with resident surveys	Active community involvement significantly strengthens environmental sustainability programs and leads to improved quality of life for locals.
De Leon, Dorig, Gozun (2024)	Profitability and Sustainability of Farm Tourism Business in Angeles City and Bacolor,Pampanga	Economic analysis and qualitative assessment	Farm tourism enterprises can be both profitable and sustainable when integrated with ecofriendly practices and supported by local policies.

Sarinas, Andrada II, Bagarinao (2023)	Gap Analysis of Sustainable Tourism: The Case of an LGU-Managed Ecotourism Project of Rizal, Laguna	Gap analysis against ecotourism criteria	There are substantial gaps in policy, implementation, and community integration that must be addressed for true sustainable tourism development in LGU-led projects.
Manalo (2017)	Relevance of Sustainable Tourism in the Philippines: Case of Boracay Island and Puerto Princesa City	Qualitative analysis against ecotourism criteria	Sustainable tourism is essential for the long-term success of major tourist destinations, necessitating stricter environmental regulations and community-based governance.
Merle U. Ruiz, Mandirigma de Vero (2019)	Strategic Marketing Framework of Community-Based Ecotourism: A case of Panguil Laguna, Philippines towards Sustainable Perspective	Mixed methods (surveys, focused groups, participatory action research)	Effective marketing strategies rooted in community empowerment can significantly enhance the sustainability of ecotourism ventures.
Mandigma Jr.,Ruiz, and De Vero (2019)	Destination Competitiveness of Beach Tourism: A case of Batangas Province Towards Sustainability	Qualitative Analysis	Boosting the competitiveness of beach destinations must balance tourist satisfaction with preserving natural resources for future sustainability.
Sandhu and Berse (2022)	A System of Systems Approach to Managing Coastal Cities: Prospects for Enhancing Sustainable Tourism and Disaster Risk Management in the Philippines	System Theory Application	Managing tourism in coastal cities requires integrated approaches linking tourism with disaster risk management and environmental sustainability frameworks.
Buhalis, D., & Amaranggana, A. (2015)	Smart Tourism Destinations Enhancing Tourism Experience Through Personalisation of Services	Conceptual framework with case studies.	Smart tourism technologies that personalize services significantly enhance tourist satisfaction and destination loyalty among visitors.

Chakrabarty, S., & Dey, S. (2020)	The Concept of Tourism Circuit Development: A Review of Literature	Literature review and analysis.	Tourism circuits are critical for connecting destinations, encouraging balanced tourism flow and regional economic growth.
Shukla, A., & Singh, A. (2024)	Exploring Sustainable Tourism Development and Visitor Experience Enhancement Through Innovative Circuit Planning and GIS Integration: A Case Study of Gorakhpur District, Uttar Pradesh, India	GIS mapping, case study, and spatial analysis.	GIS-based planning enhances tourism sustainability by linking attractions and promoting rural development.
Raj, R., & Griffin, K. (Eds.) (2017)	Cultural, Religious and Heritage Tourism in India: Tourism, Pilgrimage, and Diaspora	Edited volume with case studies.	Explores the role of cultural and religious tourism in driving tourism growth, particularly in rural and remote areas.
UNWTO (2019)	Tourism for Sustainable Development in Least Developed Countries	Global report with recommendations.	The report highlights the role of sustainable tourism in the development of least developed countries and emphasizes the importance of community-driven approaches.
Kumar, V., & Goel, S. (2017)	Tourism Circuit Planning in India: A Study on Development of Heritage Circuits	Case study analysis.	Focuses on heritage tourism circuit development and its role in preserving cultural heritage while boosting regional economies.
Sharma, A. (2018)	Tourism Circuit and Its Role in Tourism Development: An Indian Perspective	Global report with recommendations.	Highlights the significant role of tourism circuits in regional tourism development and improving local economies, with a focus on sustainability
Gretzel, U., Zhong, L., et al. (2015)	Personalization of tourism experiences: Machine learning applications in tourism circuits	Literature review & Machine Learning algorithms.	Machine learning can personalize travel experiences by suggesting destinations and activities based on past behavior and preferences.

Gretzel, U., Zhong, L., et al. (2016)	Machine learning techniques for tourism circuit recommendations	Application of machine learning algorithms for personalized recommendation systems.	ML improves tourism circuit recommendations by analyzing user preferences and enhancing destination personalization.
Mishra, R., et al. (2021)	Machine learning in tourism forecasting: Predicting tourism patterns and resource management	Machine learning forecasting models for predicting tourist behavior.	ML-based forecasting helps optimize tourism management by predicting tourist demand and patterns.
Sigala, M. (2018)	The role of machine learning and natural language processing in improving tourism services	Analysis of NLP techniques, including chatbots, and ML models in tourism.	ML and NLP improve tourist services, such as chatbots providing assistance and booking services.
Khan, M., et al. (2020)	Image recognition in tourism: Enhancing user-generated content and travel engagement	Machine learning-based image recognition and tagging for tourism platforms.	ML in image recognition increases user engagement and curates travel content based on uploaded images.
Baggio, R., et al. (2020)	Geospatial data analysis in tourism management using machine learning.	Geospatial data analysis using ML techniques (K-means clustering, Decision Trees).	ML optimizes geospatial analysis for tourism circuit planning, helping identify travel patterns.
Xu, F., et al. (2021)	Tourism Segmentation with machine learning: Targeted marketing for eco-tourist and cultural travelers	Machine learning segmentation methods, including clustering.	ML techniques segment tourists based on behaviors and preferences for targeted marketing.
Yang, W., et al. (2015)	Enhancing tourism demand forecasting using real-time internet search data	Incorporation of internet data into forecasting models	Using search data (Google Trends) improves forecasting accuracy for tourism demand.

Badouch, N., & Boutaounte, S. (2023)	Personalized travel recommendation systems and machine learning: A review	Literature review on TRS, including collaborative filtering and hybrid models.	Personalized recommendation systems based on machine learning improve travel experience and business revenues.
Perles-Ribes, J. F., et al. (2020)	Predicting overtourism using machine learning: A study of Spanish cities	ML-based analysis of tourism pressure indicators and overtourism prediction	.ML identifies over tourism risks and enables better management strategies to balance tourism pressure.
Bravo, D., et al. (2023)	Machine learning for post-COVID-19 tourism recovery: Predicting visitor numbers to the Moche Route in Peru	Use of machine learning algorithms (linear regression, KNN, decision trees).	ML models improve the accuracy of tourism forecasting, supporting tourism recovery.
Li, X. (2022)	Predicting tourism patterns in the Moche Route using machine learning: A digital transformation case study	Use of SPCA-CNN-LSTM model for tourism demand forecasting	.ML improves visitor prediction accuracy, aiding tourism management and sustainable development.
Rubi, M., et al. (2022)	Predicting travel insurance purchases using machine learning: An analysis of consumer behavior	Classification algorithms (Random Forest, Decision Tree, Stochastic Gradient Descent).	ML algorithms predict travel insurance purchases based on demographic data.
Núñez, J., Gómez-Pulido, J. A., & Ramírez, J. (2024).	Machine Learning Applied to Tourism: A Systematic Review.	Systematic Literature Review (SLR) of peer-reviewed articles using bibliometric analysis and classification by tourism domain, ML technique, and problem type.	Supervised Learning (SL) techniques, especially neural networks, are predominant in tourism applications; Key application areas include demand forecasting, personalization, sentiment analysis, and operational optimization. ;Emerging use of Deep Learning (DL) and hybrid models.

Conceptual Framework

Figure 1 presents the conceptual framework for developing a web-based, machine learning-integrated tourism circuit platform for the Municipality of Majayjay, Laguna. The framework begins with the Agile Scrum methodology, which ensures iterative development through sprints, continuous feedback, and adaptive planning. The system is built on a modular architecture that includes Account Management, Tourist Listings, Accommodation Management, a Machine Learning Recommendation Engine, and an Admin Dashboard, enabling user interaction, content management, and personalized suggestions. It also incorporates Geographic Information System (GIS) technologies for spatial data processing, route visualization, and interactive mapping, supported by API integration and GIS interfaces. For evaluation, the framework applies the ISO 25010 software quality standard and the Technology Acceptance Model (TAM) to assess system usability and user acceptance. Overall, this framework integrates agile development, modular design, GIS, and evaluation strategies to deliver a smart tourism platform that promotes sustainability and enhances the visitor experience in Majayjay, Laguna.

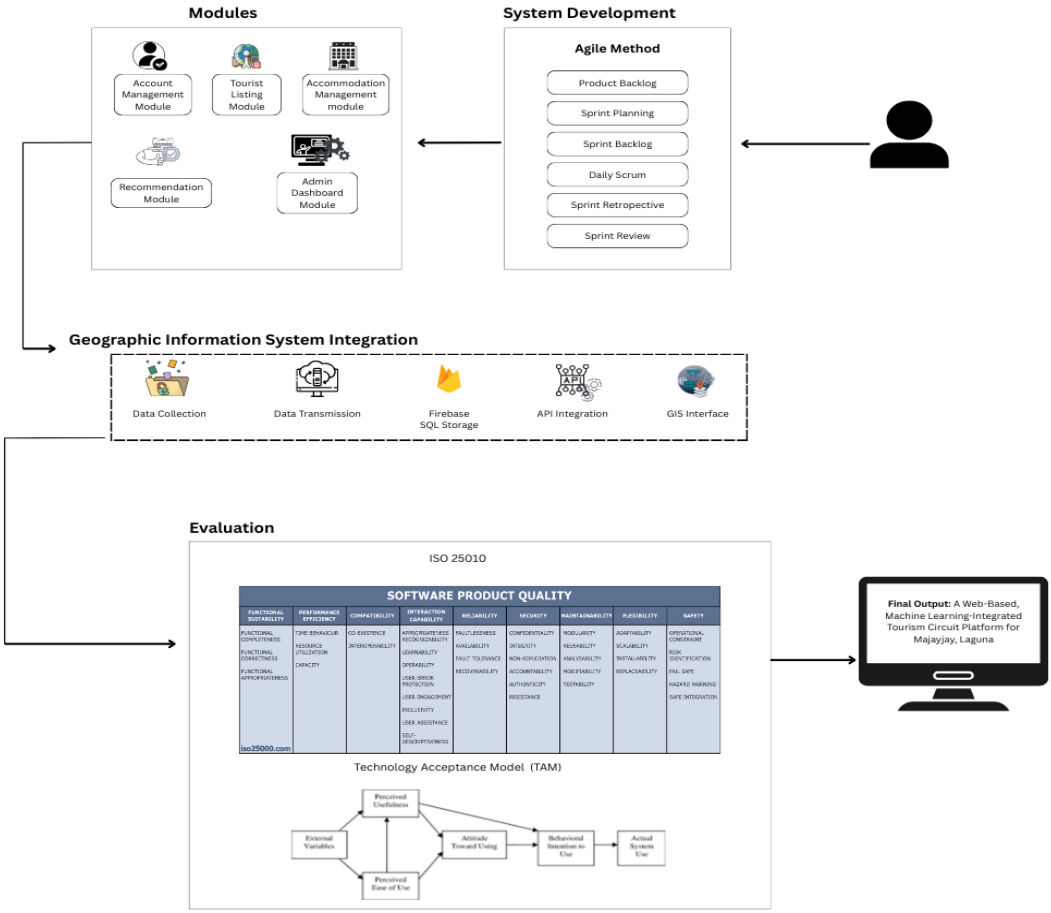


Figure 1. Conceptual Framework of the Web-Based Tourism Circuit Platform for Majayjay, Laguna

CHAPTER 3

METHODOLOGY

Project Design

This section introduces the overall design and architecture of the proposed web-based tourism platform for Majayjay, Laguna. It outlines the system structure, major components, and core functionalities. The project follows a modular and layered design approach to ensure scalability, performance, and ease of maintenance. The system integrates machine learning for personalized tourist experiences and employs a client-server model to manage interactions between users, server processes, and the database.

Figure 2 shows the system structure of the Web-Based System. It is composed of three layers; User Interface, System Modules and the Database. The system modules contain five modules which are the account management module, tourist listing module, accommodation module, recommendation module and the admin dashboard.

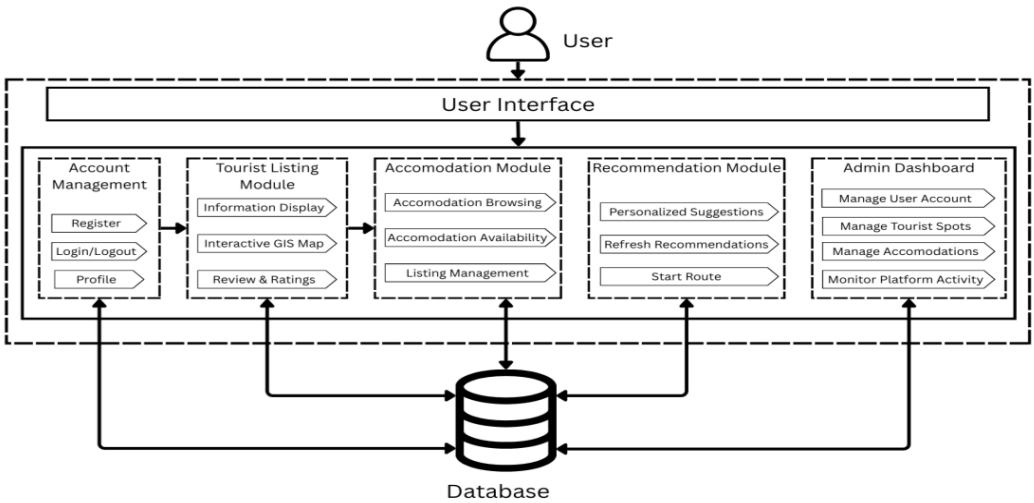


Figure 2. System Structure of the Web-Based System

Software Development

Figure 3 illustrates the Agile Scrum methodology used in the development of the platform. Agile Scrum was chosen because the project requires adaptability, regular delivery of working features, and the ability to quickly respond to feedback from users and stakeholders.

This approach also allows the team to identify and address problems early in the development process, reducing the risk of major issues later on. It helps the team manage risks more effectively by breaking the project into smaller, manageable sprints, where progress can be reviewed and adjusted regularly. Each sprint is designed to deliver real and usable features, ensuring that continuous value is provided to the users. By using Agile Scrum, the development process remains flexible and efficient, allowing to adapt to changes in requirements and feedback.

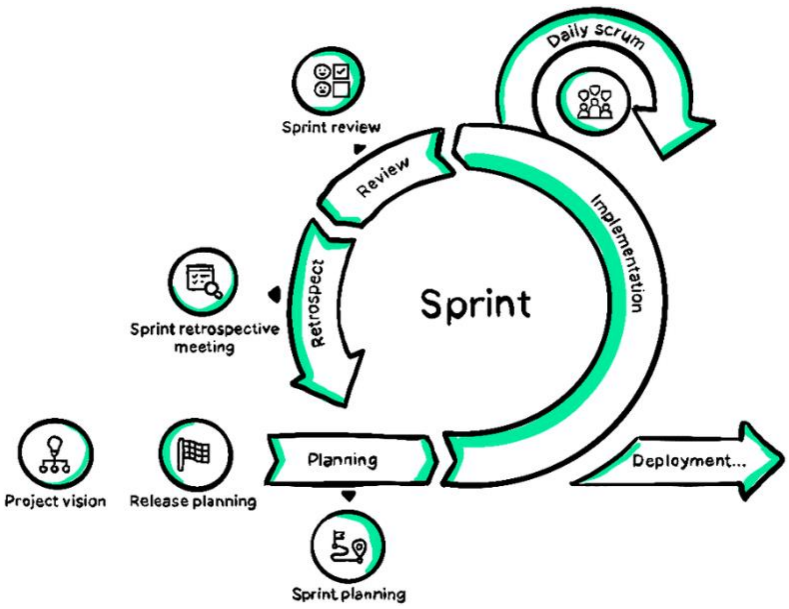


Figure 3. Agile Scrum Methodology

Product Backlog

The Product Backlog is a prioritized list of tasks, features, and requirements that need to be completed for the development of the Web-Based Tourism Platform. It serves as the foundation for the project, capturing everything from technical requirements to user stories, and ensures that the development team is focused on delivering the most valuable and essential features first.

Table 2 highlights the functionalities available for the Tourist user, presenting the key features that allow tourists to interact with the system in a simple and convenient manner. The table outlines the essential tasks that tourists can perform, starting from account registration and login, which enable access to personalized services and saved preferences. Tourists can search for tourist spots and accommodations based on their interests, location, price range, and ratings, helping them find destinations that best suit their needs. The system also allows users to view detailed information about each tourist spot, including images, descriptions, and ratings, which supports better decision-making before visiting. Through the platform, tourists can book accommodations directly, making trip planning more efficient. In addition, the Machine Learning–based recommendation feature provides personalized suggestions to guide tourists toward attractions they may enjoy. The integrated GIS tourism map further enhances exploration by allowing users to easily view nearby tourist spots and plan their routes. After completing their trips, tourists can leave reviews and feedback, contributing to the community and helping future visitors

Table 2. Product Backlog of Tourist

Priority	Feature	Est. Hour/Day
1	As a Tourist, I want to sign up and log in so I can access personalized services.	4 hrs
2	As a Tourist, I want to search for tourist spots and accommodations based on location and interest.	6 hrs
3	As a Tourist, I want to view detailed information (images, description, rating) about a tourism spot.	5 hrs
4	As a Tourist, I want to book an accommodation through the platform.	6 hrs
5	As a Tourist, I want to receive recommendations for tourism spots.	8 hrs
6	As a Tourist, I want to view a GIS map to explore nearby tourism spots easily.	7 hrs
7	As a Tourist, I want to leave a review after visiting an accommodation or spot.	4 hrs
8	As a Tourist, I want to filter search results by price, rating, or distance.	6 hrs
9	As a Tourist, I want to recover my password if I forget it.	3 hrs

Table 3 focuses on the Administrator side of the platform. It lists all the important actions that an Administrator must perform to keep the system functioning properly. This includes tasks such as logging into the system, editing tourist spot information, managing user accounts, generating reports, and handling payments. The Administrator has full control over the system to ensure that all users are legitimate, tourist information is up to date. This table illustrates the critical administrative responsibilities needed for platform governance, system maintenance, and quality assurance.

Table 3. Product Backlog of Administrator

Priority	Features	Est. Hour/Day
1	As an Administrator, I want to log into the system so that I can manage users and content.	3 hrs
2	As an Administrator, I want to add, edit or remove tourism spot information to keep it updated.	5 hrs
3	As an Administrator, I want to add, edit or remove accommodation information to keep it updated.	6 hrs
4	As an Administrator, I want to manage users (suspend/delete) in case of violations.	5 hrs

Sprint Planning

Sprint planning is a crucial part of the Agile development process where the project team outlines the tasks and goals for the upcoming sprint. During sprint planning, the team discusses and prioritizes the features, tasks, and deliverables that need to be completed within the sprint, which typically lasts from one to four weeks. The tasks are broken down into manageable pieces, and the team estimates the time required to complete each one. The main goal of sprint planning is to ensure that everyone understands what needs to be done, set realistic expectations, and align efforts toward delivering a functional and high-quality product at the end of the sprint.

System Flowchart

The system flowchart shows how users interact with the platform. It includes steps like signing up, browsing destinations and booking stays. It helps visualize how the system works from start to finish.

Figure 4 Shows that the tourist begins by registering or logging into the platform, where they gain access to personalized features. Once logged in, they can browse available accommodations and explore various tourist spots in Majayjay, Laguna. The system allows them to search and filter options based on specific criteria like location, price, and amenities. Additionally, the machine learning recommendation system suggests personalized tourist spots and accommodations based on the tourist's preferences and past behavior. Once they find suitable options, the tourist can proceed to make a booking through the platform.

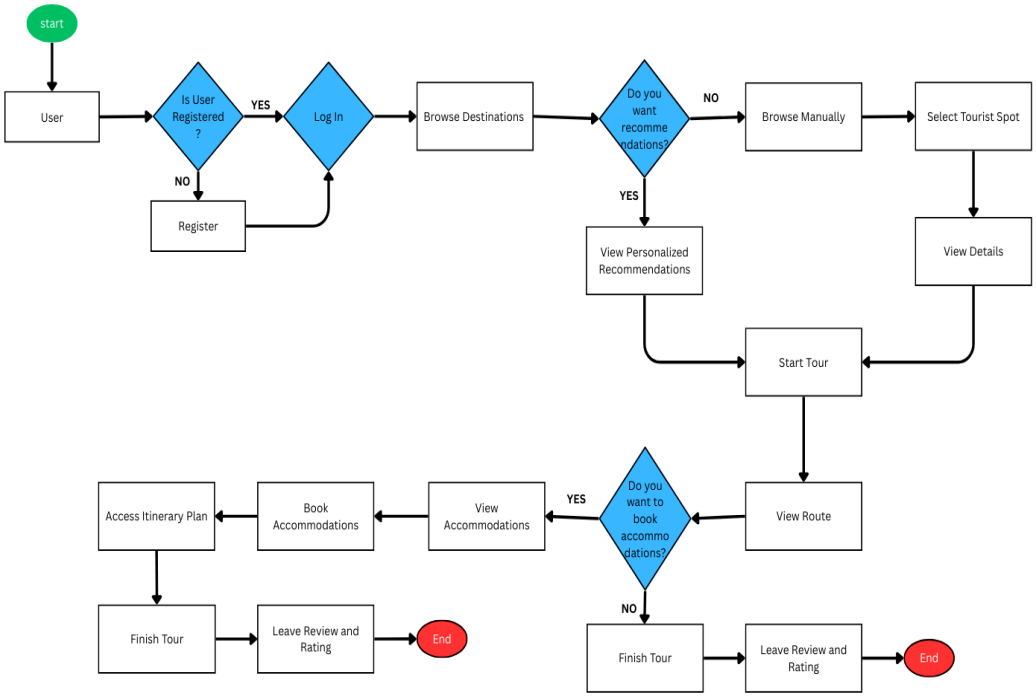


Figure 4. Flowchart of Tourist

In Figure 5 The admin accesses the platform by logging in to the admin panel. Through this panel, they have the ability to monitor the entire system, ensuring that everything runs smoothly. The admin can generate detailed reports related to user activity and overall platform performance. This helps in making data-driven decisions, maintaining system health, and ensuring that both tourists and hosts are receiving the service they expect.

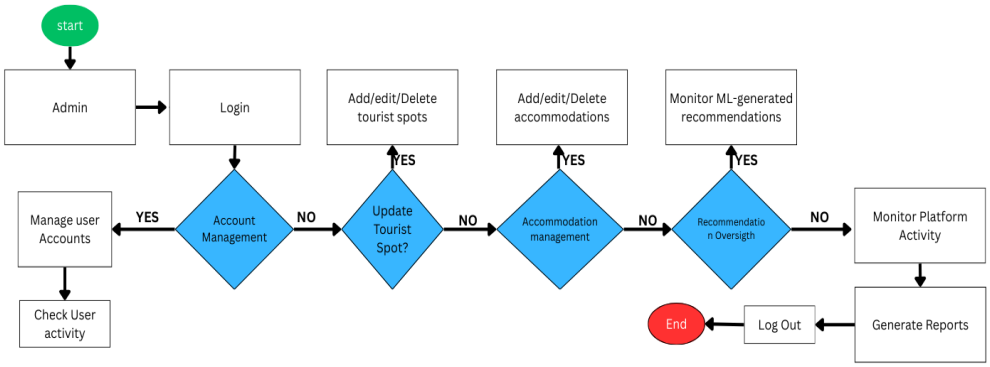


Figure 5. Flowchart of Admin

Use Case Diagram

Figure 6 illustrates the different users of the system and the actions they can perform. The Tourist user can browse available tourist spots and accommodations, view detailed information, and rate or provide feedback based on their experiences. These actions allow tourists to actively interact with the platform and make informed travel decisions. On the other hand, the Admin is responsible for overseeing the entire system. The Admin manages system content, monitors user activity, and ensures that all information remains accurate and up to date.

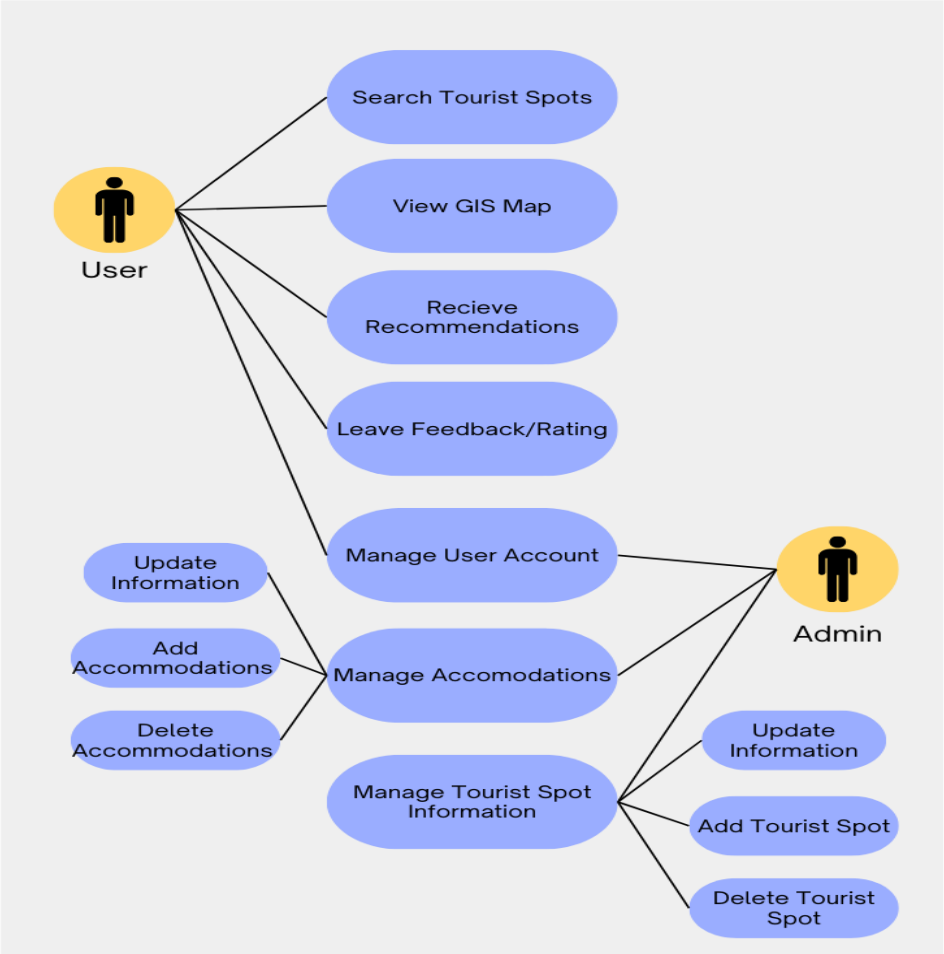


Figure 6. Use Case Diagram

Sprint Backlog

Table 4 outlines the key components for the platform's success, prioritizing core modules like User Registration, Accommodation Listings, GIS Mapping, and Machine Learning Recommendations. These were scheduled early to build a solid foundation before adding features like Search/Filter and Feedback systems. Admin functions follow after main user features. Sprint planning considered time for coding, design, debugging, and testing, assigning timelines based on module complexity. Complex tasks like GIS and machine learning were given more time.

Table 4. Sprint Backlog

Priority	Features	Task	Est. Hour/Day
1	User Registration/Login	Design and Implement registration/Login Form	5 hrs
2	User Profile Management	Create user profile edit and view pages	5 hrs
3	Accommodation Listings	Develop accommodation listing module	8 hrs
4	GIS Mapping Integration	Integrate tourist spot maps using GIS	8 hrs
5	Machine Learning Recommendation	Set up basic ML model for personalized tours	8 hrs
6	Search and Filter Functionality	Implement search by location/amenities	6 hrs
7	Feedback and Rating System	Develop review and rating functionality	14 hrs
8	Admin Panel	Create dashboard for site management	16 hrs
9	User Activity Tracking	Implement basic activity logs	16 hrs

Materials

This section talks about the specific specifications of both the hardware and software that was used in the development of the system. Providing an overview of resources to create this project.

Software

Listed below are the list of software used during the development phase of the web-based system.

React.js was the main tool we used for building the system's interface. It allowed us to design pages that are interactive, easy to use, and responsive on different devices.

Node.js was used to connect the different parts of th`e system and make sure everything worked together properly. It served as the middle layer that handled requests, such as saving bookings, retrieving tourist spot details, and passing information between the website and the database. Its reliability made it suitable for managing the platform's activities smoothly.

Flask (Python) supported the machine learning component of our project. We used it to run our recommendation system, which suggests tourist spots based on user preferences and behaviors. Flask made it easier to connect our machine learning models with the main website so users could receive personalized recommendations in real time.

XAMPP and MySQL were used for storing and managing important information in the system. XAMPP provided a local environment where we could test the website before deployment, while MySQL served as the database where we kept records such as user accounts, tourist attractions, accommodations, and bookings. These tools were free, practical, and reliable for our needs.

Leaflet.js was used to display interactive maps in the platform. With this tool, we were able to show the locations of tourist spots, accommodations, and other points of interest in Majayjay, Laguna. It would users to easily explore the area visually and plan their trips more effectively.

Hardware

The hardware used in the development, testing, and implementation is listed is listed in this part.

Table 5 shows the minimum hardware specifications required for the System to run efficiently. These specifications ensure that the system can handle the basic tasks of user registration, accommodation listings, payment processing, and machine learning recommendations. The Intel Core i5 processor, 4GB of RAM, and Windows 10 operating system are necessary to provide smooth performance, while the 56GB of storage is sufficient for system files and user data. Meeting these minimum requirements would allow users to experience the core functionalities of the platform without performance issues.

Table 5. Hardware Specifications

Minimum Requirements	Specification
CPU	Intel Core i5 Processor
RAM	4GB RAM
OS	Windows 10
Storage	56GB

Data Collection

Data would be gathered from various sources to build a recommendation system. User demographics like age and preferences would be collected through surveys and registration. Tourist behavior data, including ratings, would help understand user interests. Information on tourist spots, accommodations, and local businesses would be gathered to provide relevant recommendations.

Data from open sources like local government units and online review platforms would also be used for updated information.

Data Pre-processing

After data collection, the data would be cleaned by removing duplicates and handling missing values. It would be transformed into a usable format, such as

encoding categorical data and normalizing numerical data. Feature engineering would create additional features, and the data would be split into training, validation, and testing sets for model training and evaluation.

Model Development

The recommendation system uses a rule-based learning approach that calculates the cosine similarity to generate personalized travel suggestions. The model uses predefined rules that evaluate multiple factors influencing a tourist's preferences, including user feedback, seasonal patterns, time-based preferences, weather conditions, special occasions, and user location. These rules determine which destinations are relevant under specific circumstances.

To further personalize recommendations, each point of interest (POI) is represented as a feature vector, combining enhanced categorical features, tag-based features, visitors popularity scores, and TF-IDF features of the POI descriptions. The system computes cosine similarity between a simulated user profile and each POI's feature vector, producing a base similarity score. This score is then adjusted by the context-based boost factors derived from the rule-based layer, resulting in a final recommendation score for each POI.

Figure 7 below illustrates how the system generates personalized suggestions by analyzing user feedback. It shows that the platform considers users' liked destinations to recommend tourist spots that match their interests, providing a more tailored travel experience.

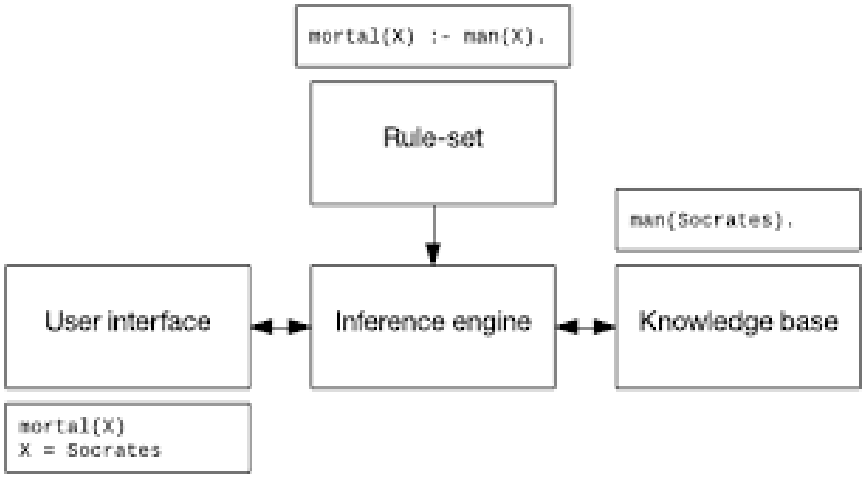


Figure 7. Rule-Based Learning

Model Evaluation

The Majayjay Tourism recommendation system suggests tourist spots based on how similar each place is to a user's preferences. The system uses cosine similarity, which measures how closely a user profile matches the features of each point of interest (POI). Each POI is represented by a combination of categories (like adventure, dining, or water activities), tags, visitor popularity scores, and the text description processed using TF-IDF.

The system evaluates its performance by generating a list of top recommendations and comparing them with user feedback. Metrics such as **precision** measure the proportion of recommended tourist spots that users actually find relevant, while **recall** determines how many of the user's preferred

or relevant spots were successfully included in the recommendations. The **F1-score** is then used to provide a balanced evaluation by combining both precision and recall, ensuring that the recommendation model is assessed for both accuracy and completeness.

Figure 8 shows how the system calculates similarity using the cosine formula. Each tourist spot is converted into a feature vector that includes numerical values, tag indicators, enhanced categories, and TF-IDF description features. The system calculates the cosine similarity between a user profile vector and each POI vector to get base similarity scores. These scores are then adjusted using temporal and contextual boosts, such as time of day, season, and weather, producing the final ranked list of recommendations.

$$similarity(A,B) = cos(\theta) = \frac{A \cdot B}{||A||\,||B||} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2 \sum_{i=1}^{n} B_i^2}}.$$

Figure 8. Cosine Similarity Algorithm

Project Testing and Evaluation

The project would undergo thorough testing after development to ensure it meets all functional and non-functional requirements. This would include unit testing, integration testing, compatibility testing, and system testing to verify that the system functions correctly, works well across different platforms, and performs as expected as a whole.

Unit Testing

The project would undergo unit testing to verify that each individual component of the system functions correctly as intended. Each module, such as user registration, login, booking, and recommendation features, would be tested independently to identify and fix errors at an early stage of development.

Integration Testing

Integration testing would be conducted to ensure that the different modules of the system work together properly. This testing would focus on data flow and interaction between components, such as database connections, user interfaces, and system features, to confirm that integrated modules operate smoothly and without conflicts.

Compatibility Testing

The system would be tested across a range of devices, including desktops, laptops, tablets, and smartphones, to ensure it displays and operates properly on different screen sizes and resolutions. The system would be tested on major browsers such as Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge to verify correct layout, functionality, and responsiveness.

System Testing

The researchers would test all the features of the system based on the objective. The researchers would do a full system testing to verify that all modules, features, and integrations function correctly and efficiently.

Evaluation Procedure

The researchers used the Technology Acceptance Model (TAM) to evaluate the system, focusing on perceived usefulness and ease of use. Evaluation forms based on TAM were given to tourists, and IT experts to gather feedback on system navigation, usefulness, and satisfaction. The results helped assess the platform's strengths, weaknesses, and areas for improvement.

ISO

The ISO 25010 model was applied in this study to evaluate the overall quality of the system. ISO 25010 focuses on eight key quality characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Figure 9 illustrates the ISO 25010 quality model, ensuring that the tourism platform is useful, dependable, efficient, and safe forusers.

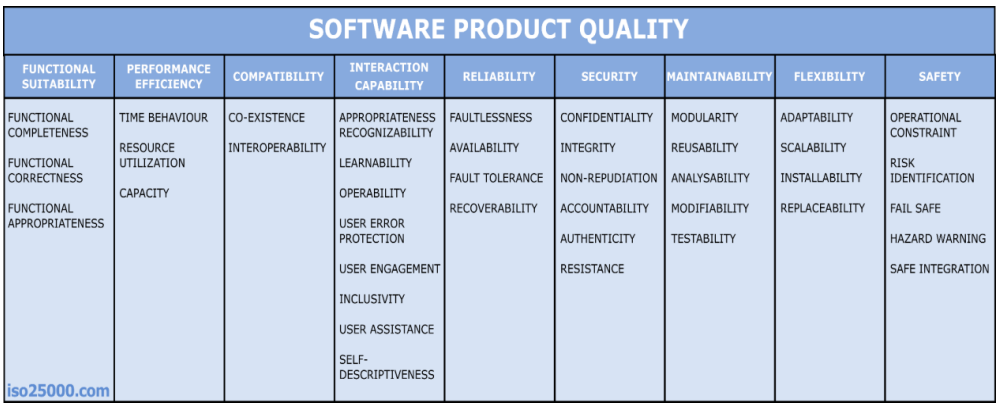


Figure 9. Product Quality Model

Technology Acceptance Model

Figure 10 shows the Technology Acceptance Model (TAM). It gives a clear and simple way to understand how users accept and use a new system. In this study, TAM was used to evaluate the Web-Based Tourism Platform for Majayjay, Laguna. It focuses on five key parts: External Variables, Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Attitude Toward Using (A), Behavioral Intention to Use (BI), and Actual System Use.

External Variables look at outside influences like past experiences. PU and PEOU measure how helpful and easy the system is. Attitude (A) shows the users' feelings about the system. BI measures their plan to keep using it, and Actual System Use tracks real usage.

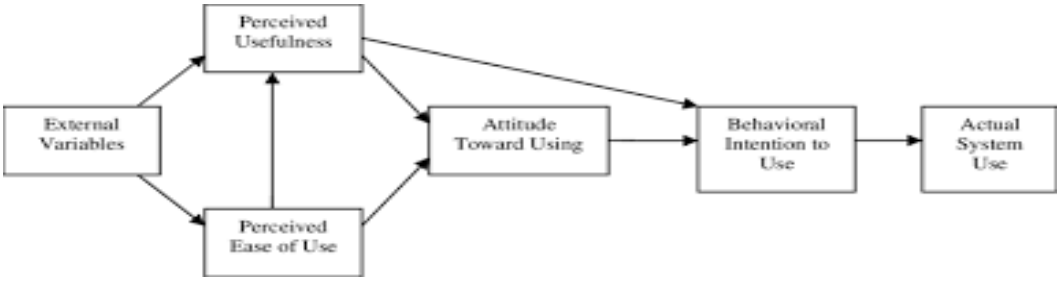


Figure 10. TAM

Sampling Design

The researchers used purposive sampling, a technique in which participants are selected based on specific characteristics relevant to the study. The respondents included tourists who have visited or not visited Majayjay, Laguna, and IT experts who provided technical insights on the platform’s design, usability, and functionality.

Population

The researchers would select 100 respondents and three IT experts to participate in the study. The respondents would be guided during the testing process to ensure proper system usage and to evaluate its functionality and usability, while the IT experts would focus on assessing data security, system performance, and overall technical reliability.

Table 6. Respondent of the study

Type of Respondents	Number of Respondents
User	100
IT Expert	3

Data Collection Instrument

The questionnaires and surveys used for data collection were developed based on the Technology Acceptance Model (TAM), which is commonly used to measure how users accept and use a system. The questions focused on important TAM elements adapted from Diaz et al., such as how useful and easy to use the system is, to ensure consistency in evaluating user acceptance. These questions were designed to clearly understand the users' opinions and experiences while using the platform.

To ensure that the questionnaire was reliable, Cronbach's Alpha was applied to check whether the questions consistently measured the intended factors. The results showed acceptable Cronbach's Alpha values, indicating that the questionnaire is reliable and that the survey tool is appropriate for evaluating user acceptance and the overall effectiveness of the system.

Table 7 shows the reliability results of the survey items based on the Technology Acceptance Model (TAM). Each construct—such as Perceived Usefulness, Perceived Ease of Use, Perceived Cost, and Wouldingness to Adopt—was measured using 3 to 4 items. Cronbach's Alpha values were calculated to check internal consistency. Results show that most constructs achieved an "Excellent" rating (above 0.90), while others were rated "Good" (0.80–0.89). This confirms that the questionnaire is reliable for assessing user acceptance of the system.

Table 7. Reliability result for Technology Acceptance Model

Item	Number of Items	Cronbach's	Interpretation
Perceived Usefulness (PU)	4	0.935	Excellent
Perceived Ease of Use (PEOU)	4	0.956	Excellent
Perceived Cost (PC)	4	0.856	Good
Willingness to Adopt (WTA)	3	0.845	Good

Legend above 0.90 = Excellent, 0.80 - 0.89 = Good, 0.70 - 0.79 = Acceptable, 0.60 - 0.69 = Questionable, 0.50 - 0.59 = Poor, Below 0.50 = Unacceptable

Statistical Treatment

To evaluate and analyze the information gathered by the researcher, the Likert Scale Method was used. The formula for calculating each result is shown in the equation below.

Table 8 shows the interpretation and representation of each of the data gathered by the researchers through the help of the questionnaires. The values shown on the table in which, 1 Very Unsatisfied is the lowest value with range of 1.00 - 1.80 while 5 Very Satisfied is the highest value attainable with the range of 4.21- 5.00.

Table 8. Likert Scale Rating

Rating	Range	Interpretation
5	4.21 - 5.00	Very Satisfied
4	3.41 - 4.20	Satisfied
3	2.61 - 3.40	Neutral
2	1.81 - 2.60	Unsatisfied
1	1.00 - 1.80	Very Unsatisfied

Ethical Consideration

Ethical considerations would focus on protecting user data, promoting transparency, ensuring fairness, and supporting sustainable tourism. The system would collect only necessary information with users' full consent, and clear privacy policies would be provided. Security measures such as encryption and access controls would safeguard user information. Data gathered from surveys and interviews would also be handled with strict confidentiality.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the result and discussion to support the methodology utilized in the achievement of the objectives.

Result By Objectives

This section presents the system outputs interface designs and evaluation results based on the specific objectives of the study.

Majayjay Tourism Circuit Recommendation System

Figure 11 below shows the main interface of the Majayjay Tourism Circuit Web Platform, highlighting its core features designed to help tourists explore Majayjay's destinations easily. The interface presents navigation options for Points of Interest (POIs), lodging listings, the user dashboard, and itinerary planning. The page layout displays essential tourism data in an organized and intuitive manner, allowing users to browse destinations and accommodations efficiently.

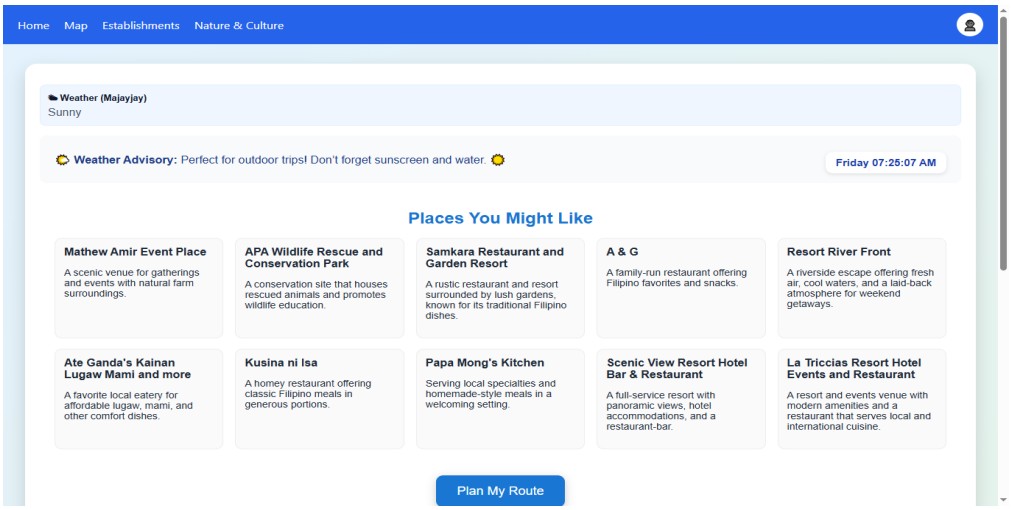
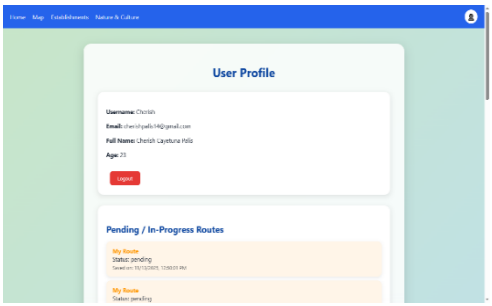
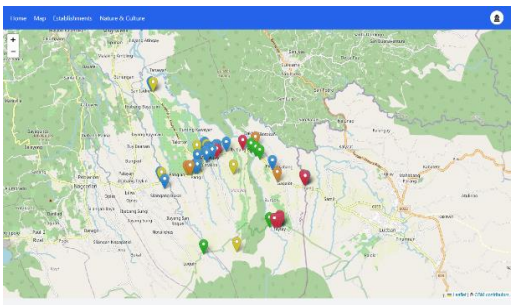


Figure 11. User Interface

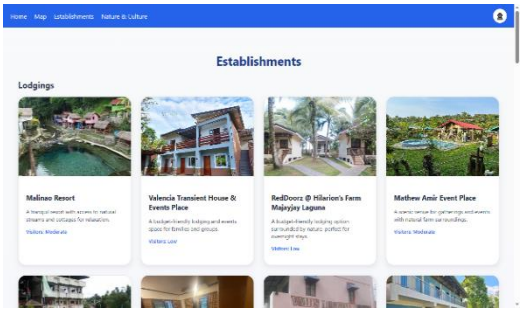
Figure 12 illustrates the main system modules. Section (a) User Account Module allows users to register, log in, and manage their profiles securely, including saving routes and preferences through token-based authentication. Section (b) Tourism Listings Module presents Majayjay's major attractions with essential details such as descriptions, images, coordinates, and map-based visualization. Section (c) Lodging Information Module provides accommodation details including price range, amenities, and proximity to tourist spots to help users choose suitable lodging. Section (d) Administrative Dashboard enables administrators to manage tourist spots, lodging information, user accounts, and system reports to ensure smooth platform operation.



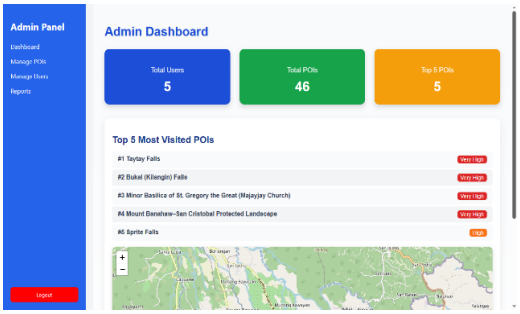
a



b



c



d

Figure 12. User Interface for recommendation system (*a*) Account Module, (*b*) Tourist Listing Module, (*c*) Lodging Information, and (*d*) Admin Dashboard

Unit testing result

Table 9 shows the unit testing for the Account Module functions and their expected output, actual output, and response time. The remarks indicate that the Account Module is functional, as all test cases passed the unit testing. However, the response time for each function varies due to factors such as internet speed, server load, and the size of data being processed. After corrections, all functions performed as expected and passed the unit testing.

Table 9. Account Module Unit Tests

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT / RESPONSE TIME	REMARKS
Valid user registration	Success message; user added to database	User registered successfully (1–2 seconds)	PASSED
Invalid email format	Error message: "Invalid email format"	Error displayed immediately	PASSED
Duplicate email registration	Error message: "Email already registered"	Error displayed immediately	PASSED
Valid login	Redirect to dashboard; session created	Login successful, session active (1–2 seconds)	PASSED
Invalid login (wrong password)	Error message: "Invalid credentials"	Error displayed immediately	PASSED
Logout functionality	Session destroyed; redirect to login page	Logout successful (1 second)	PASSED
Valid user registration	Success message; user added to database	User registered successfully (1–2 seconds)	PASSED
Invalid email format	Error message: "Invalid email format"	Error displayed immediately	PASSED
Logout functionality	Session destroyed; redirect to login page	Logout successful (1 second)	PASSED

Table 10 presents the unit testing results for the Tourist Spot Module, detailing the specific test cases conducted, the corresponding expected outputs, the actual outputs, and the remarks indicating the test status. The results demonstrate that all identified functionalities of the module were tested individually and performed as intended, with each test case successfully passing the defined criteria. This indicates that the module is fully functional and meets the system requirements. Response times varied slightly across different functions, which can be attributed to factors such as internet connectivity, server performance, and the amount of data processed. Overall, the unit testing results confirm that the module operates correctly, produces accurate and consistent outputs, and demonstrates reliable performance.

Table 10. Tourist Listing Module Unit Tests

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT / RESPONSE TIME	REMARKS
Display all tourist spots	List of all tourist spots with details	All spots displayed (2–3 seconds)	PASSED
Search tourist spot by name	Details of the searched spot displayed	Search results shown (1–2 seconds)	PASSED
Filter by category	Only spots from selected category shown	Filter applied successfully (1 second)	PASSED
Filter by like/dislike ratio	Spots meeting like threshold displayed	Filtered list displayed (1 second)	PASSED
View spot details	Detailed view with images, description, map	Details page loaded (1–2 seconds)	PASSED
Invalid search query	Message: "No results found"	No results message shown (1 second)	PASSED
Sort by popularity	List sorted by like count descending	List sorted correctly (1 second)	PASSED

Table 11 presents the unit testing results for the Accommodation Module, detailing the tested cases, expected outputs, actual outputs with response times, and corresponding remarks. The results show that all test cases passed successfully, indicating that the Accommodation Module is fully functional and complies with the specified system requirements.

Administrative functions, including adding, editing, and deleting lodging information, were executed accurately and consistently, with all changes correctly reflected in both the system database and the user interface. User-side features, including viewing lodging details, filtering by lodging type, and sorting by distance, worked as expected and responded within acceptable time frames. Variations in response time were minimal and may be influenced by factors such as server load and data processing. Overall, the testing confirms that the Accommodation Module operates reliably and efficiently.

Table 11. Accommodation Module Unit Tests

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT / RESPONSE TIME	REMARKS
Admin adds new lodging info	New lodging added to database; visible in listings	Lodging added successfully (2–3 seconds)	PASSED
Admin edits lodging info	Lodging updated in database	Details updated successfully (2 seconds)	PASSED
Admin deletes lodging info	Lodging removed from database and UI	Lodging removed (1 second)	PASSED
User views lodging details	Full details displayed (amenities, contact)	Details page loaded (1–2 seconds)	PASSED
Filter by lodging type	Only selected type displayed	Filter applied (1 second)	PASSED
Sort by distance	List sorted by nearest first	Sorting applied (1 second)	PASSED

Table 12 presents the unit testing results for the Recommendation Module, detailing the tested cases, expected outputs, actual outputs with response times, and remarks. The system was able to generate recommendations for new users, provide personalized suggestions based on user likes, and deliver context-aware and seasonal recommendations that align with time and weather conditions. The current season in which the system was tested in was in November, therefore the expected output of the recommendation module would be places that has a dry-cool atmosphere. In which the recommendation system responded with the outcome of places accordingly. Moreover, the recommendation limit feature worked correctly by displaying only the specified number of results. Response times were within acceptable ranges, indicating that the module performs efficiently and reliably

Table 12. Recommendation Module Unit Tests

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT / RESPONSE TIME	REMARKS
Recommendations for new user	Default popular spots shown	Recommendations displayed (2–3 seconds)	PASSED
Recommendations based on likes	More spots similar to liked categories shown	Personalized list displayed (2 seconds)	PASSED
Context-aware recommendations	Recommendations match time and weather	Context-aware results shown (2 seconds)	PASSED
Seasonal recommendations	Spots suitable for current season displayed	Seasonal spots shown (2 seconds)	PASSED
Recommendation limit	Only specified number of recommendations shown	Limited list displayed (1 second)	PASSED

Table 13 shows the unit testing results for the Like/Dislike Module, including the test cases, expected results, actual results, response times, and remarks. The results show that the module works correctly and meets all the requirements because all test cases passed successfully. Users can like or dislike tourist spots, change their feedback anytime from like to dislike or vice versa, and even remove it completely. The system updates all actions correctly and quickly, usually within about 1 second.

Guests can also see the total number of likes and dislikes without clicking anything, which makes it easy to know how popular a place is. Overall, the module works as intended. It allows users to share their opinions about tourist spots, change their minds if their experience changes, and see what other people think. The system handles all feedback properly and gives a fast and smooth experience, making it simple and reliable for everyone to use. This feature helps the whole system by giving users a way to interact with the platform and see community opinions.

Table 13. Like/Dislike Module Unit Tests

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT / RESPONSE TIME	REMARKS
User likes a tourist spot	Like count increments; visual feedback	Like recorded (1 second)	PASSED
User dislikes a tourist spot	Dislike count increments	Dislike recorded (1 second)	PASSED
Change like to dislike	Like removed, dislike added	Feedback updated (1 second)	PASSED
Remove feedback	Like/dislike removed	Feedback cleared (1 second)	PASSED
Guest view feedback	Counts visible, no interactive buttons	Counts displayed; buttons (1 second)	PASSED

Table 14 presents the unit testing results for the Admin Dashboard, showing the test cases, expected results, actual results, response times, and remarks. The results show that the Admin Dashboard works properly, as all test cases passed successfully. Admin login works correctly for both valid and invalid credentials, making sure that only authorized users can access the system. Administrative functions like viewing the user list, suspending or deleting accounts, and checking system logs all worked accurately and updated in the system right away.

Overall, the module gives administrators reliable tools to manage the system and keep everything under control. It also helps the whole platform run better because admins can quickly handle user accounts and monitor system activity, keeping the system safe and organized for everyone.

Table 14. Admin Dashboard Unit Tests

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT / RESPONSE TIME	REMARKS
Admin login valid credentials	Redirect to admin dashboard	Dashboard loaded (2 seconds)	PASSED
Admin login invalid credentials	Error message; no access	Access denied message shown (1 second)	PASSED
View user list	List of all registered users	User list displayed (2 seconds)	PASSED
Suspend user account	User account disabled; cannot login	Account suspended (1 second)	PASSED
Delete user account	User removed from database	User deleted (1 second)	PASSED
View system logs	Logs displayed with timestamps and actions	Logs loaded (2–3 seconds)	PASSED

Table 15 presents the unit testing results for the GIS/Map Module, including the tested cases, expected outputs, actual outputs with response times, and remarks. The results indicate that the module is fully functional, The map loads correctly with tourist spot markers, and clicking on a marker displays detailed information instantly. Route generation between points works accurately and zooming in or out is smooth and responsive. Overall, the GIS/Map Module provides reliable performance, accurate mapping features, and a responsive user experience.

Table 15. GIS/Map Module Unit Tests

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT / RESPONSE TIME	REMARKS
Load map with tourist spots	Map displayed with POI markers	Map loaded with markers (3–4 seconds)	PASSED
Click on map marker	Info window with spot details appears	Details popup shown (1 second)	PASSED
Route generation between points	Blue route line displayed on map	Route drawn on map (2 seconds)	PASSED
Zoom in/out functionality	Map zooms in/out smoothly	Zoom works smoothly (1 second)	PASSED
Current location detection	Map centers on user's current location	Location detected, map centered (2 seconds)	PASSED

Integration Testing

Table 32 presents the integration testing results of the Majayjay Tourism Circuit Web-Based System, showing the tested modules along with their expected output, actual output, response time, and remarks.

Table 16. Results of Integration Testing

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT / RESPONSE TIME	REMARKS
Account Management and Recommendation Module	User preferences and activity history should be retrieved to generate personalized recommendations after login.	User data was successfully retrieved and personalized recommendations were displayed (3–5 seconds)	PASSED
Tourist Listing and Map (GIS) Integration	Selected tourist spots should display detailed information with accurate map pin locations.	Tourist details and map visualization were displayed correctly (2–4 seconds).	PASSED
Tourist Listing and Accommodation Management Module	Nearby accommodations should be displayed when a tourist spot is selected.	Relevant accommodations were correctly linked and displayed (2–3 seconds).	PASSED
Admin Dashboard and Tourist Listing Module	Updates made by the admin to tourist spot information should reflect immediately on the user interface.	Updated destination details were displayed in real time (3–5 seconds).	PASSED
Admin Dashboard and Accommodation Management Module	Changes to accommodation listings should be visible to tourists after admin updates.	Updated accommodation details were correctly reflected on the tourist view (3–4 seconds).	PASSED
Admin Dashboard and Recommendation Module	Newly added or updated destinations should be included in recommendation results.	Recommendation results were updated to include new destinations (4–6 seconds).	PASSED

Compatibility Testing

The system performs well in Best Practices and SEO across browsers, but shows performance issues on Microsoft Edge, while working best on Google Chrome.

Table 17. Compatibility Testing of the System

Browser	Web Page	Performance	Accessibility	Best Practices	SEO
Chrome	Login page	63	96	100	90
Chrome	Admin Dashboard	68	89	100	91
Chrome	Admin Manage POI	68	95	100	91
Chrome	Admin Manage User	78	88	100	90
Chrome	Admin Reports	78	79	100	90
Chrome	User Plan Router	78	98	96	91
Microsoft Edge	Login page	62	96	100	90
Microsoft Edge	Admin Dashboard	43	89	100	91
Microsoft Edge	Admin Manage POI	44	95	100	91
Microsoft Edge	Admin Manage User	78	88	100	90
Microsoft Edge	Admin Reports	58	79	100	90
Microsoft Edge	User Plan Router	47	98	96	91
Opera	Login page	60	100	100	90
Opera	Admin Dashboard	57	90	100	91
Opera	Admin Manage POI	48	N/A	100	91
Opera	Admin Manage User	78	92	100	90
Opera	Admin Reports	62	84	100	90
Opera	User Plan Router	56	N/A	96	91

System Testing

System testing was conducted to evaluate whether all the modules of the tourism platform work correctly and efficiently according to their intended purpose. Each module was checked to see if it produces the expected output and meets the needs of the users

Table 18 shows that all modules function correctly: the Account Module handles registration, login, and profile updates; the Tourist Listing Module displays spots with working search and filters; the Accommodation Management Module allows admins to manage listings while users view details; the Recommendation Module provides relevant suggestions; and the Admin Management Module enables efficient user and system management.

Table 18. System Testing Results

MODULE	EXPECTED OUTPUT	ACTUAL OUTPUT	REMARKS
Account Module	User should be able to register and log in; update any information about their accounts	Users were able to register, log in/out, and update profiles without issues	PASSED
Tourist Listing Module	Tourist spots are displayed with accurate information; search and filters work	Listings display correctly; search and filter functions operate as expected	PASSED
Accommodation Management Module	Admin can add, update, delete accommodations; users can view details	All CRUD operations worked as intended; details display correctly	PASSED
Recommendation Module	Personalized recommendations are generated based on user preferences	System provided relevant and useful recommendations	PASSED
Admin Management Module	Admin can manage users, manage listings.	All administrative tasks performed correctly	PASSED

Smart Recommendation Engine using Machine Learning

Figure 13 presents the Recommendation Interface of the platform, showing the system's ability to generate personalized suggestions for tourists. This interface highlights machine learning–driven destination recommendations based on user feedback and saved itineraries. When users visit their dashboard, the system automatically displays suggested spots that match their interests or past interactions.

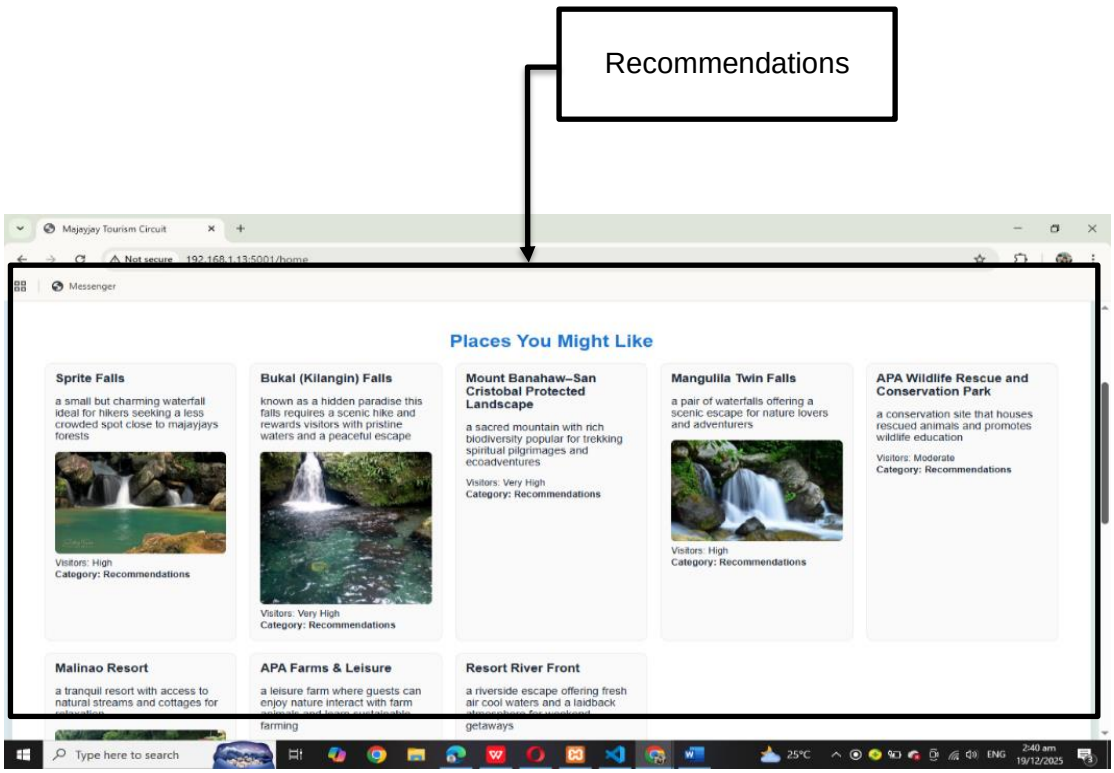


Figure 13 Recommendation Interface

Figure 14 presents the recommendation results under rainy weather conditions. This highlights how the machine learning system incorporates seasonal factors into its decision-making process, enabling it to generate more accurate, context-aware, and user-friendly recommendations.

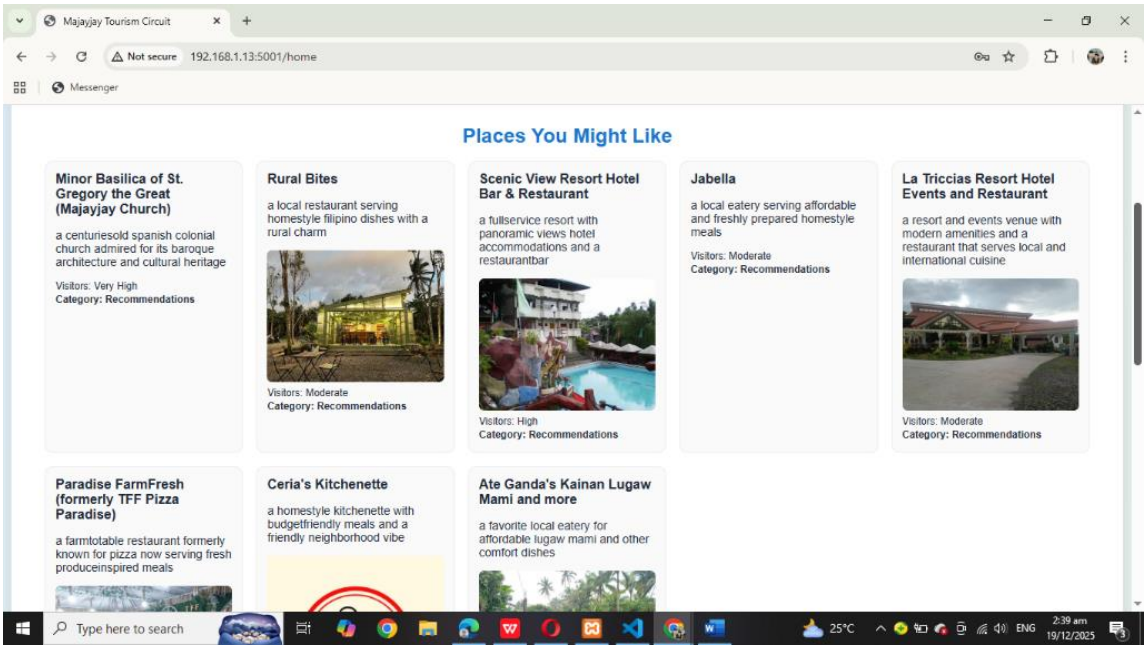


Figure 14. Smart Recommendation based on Season

Model Evaluation Result

The dataset for this study was collected from official tourism sources, field observations, and online descriptions, covering **46 tourist locations** in Majayjay, Laguna. For each location, the raw data includes the following fields: id, name, latitude, longitude, popularity, tags, and description. The tags field contains categorical attributes describing the location, such as nature, adventure, historical, or restaurant. The description field provides a textual summary highlighting key characteristics, attractions, and visitor experiences. The popularity score indicates the relative popularity of the location (Low, Moderate, High, Very High) The feature engineering process produced 16 enhanced columns and 24 tag columns for the dataset of tourist locations.

Figure 15 shows the Enhanced vs Tag Columns count. The enhanced columns were created by analyzing each location's tags, name, and description. A rule-based categorization was used to identify activity-based, facility-based, and weather-compatible categories. Each category was converted into a binary feature, resulting in 16 enhanced features that capture important characteristics of each location.

The tag columns were generated from the original tags in the dataset, where each unique tag was also converted into a binary feature. This produced 24 tag features representing different categories such as nature, adventure, historical, religious, dining, and accommodation.

Together, these features provide a complete representation of each tourist location by combining enhanced categories and original tags. This balanced feature set supports clustering, similarity analysis, and personalized recommendations in the system.

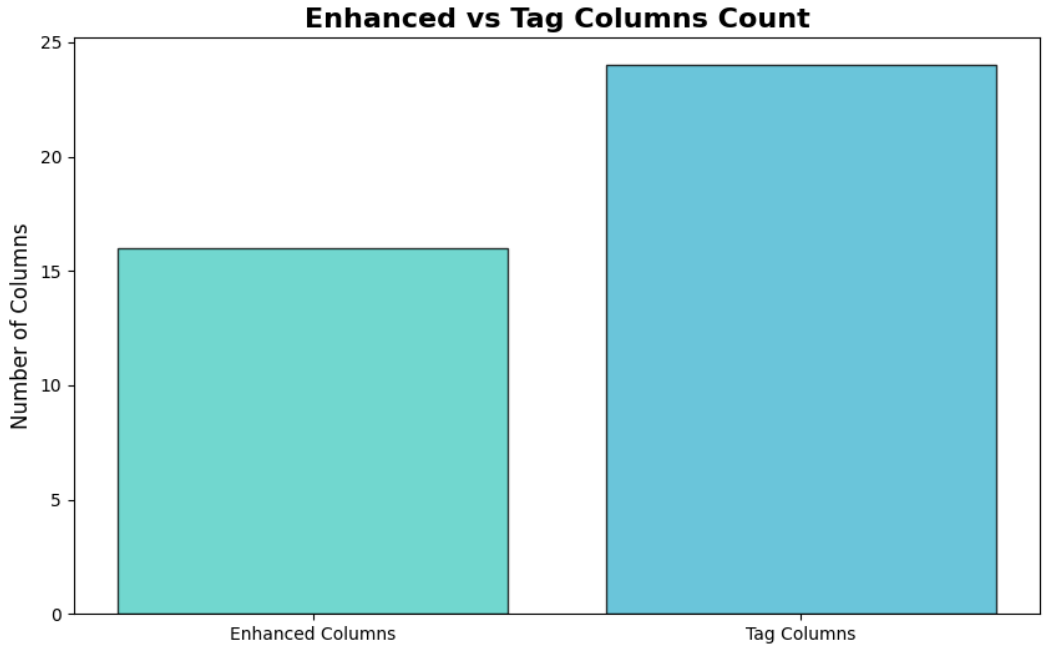


Figure 15. Enhanced vs Tag Columns Count

Figure 16 shows the feature matrix composed of 47 tourist locations, each represented by a total of 70 features. These features include quantitative attributes such as visitor ratings and scores, categorical tag features that describe the characteristics of each location, and textual features extracted from site names and descriptions using the TF-IDF technique. By combining numerical data with categorical and textual information, the feature matrix provides a detailed representation of each tourist spot.

This high-dimensional feature representation allows the recommendation system to better understand both numeric similarities, such as popularity or ratings, and semantic similarities, such as shared themes or activities between locations. As a result, the feature matrix serves as a strong foundation for clustering tourist spots, measuring similarity, and generating more accurate and personalized recommendations for users.

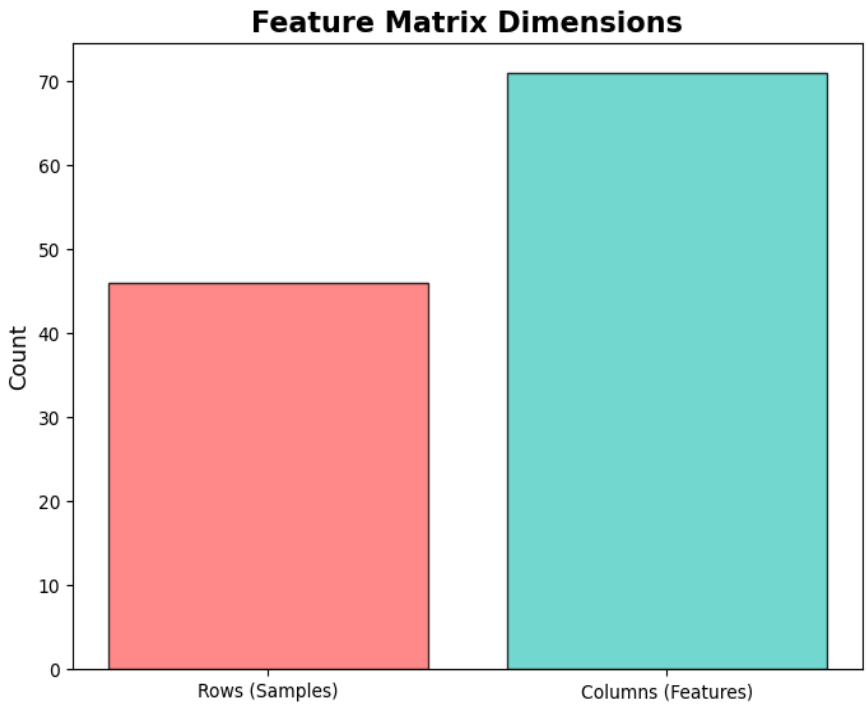


Figure 16. Feature Matrix

The recommendation scores reflect how well each location aligns with user preferences based on contextual factors such as visitor popularity, activity type, season, time of day, and weather. Sprite Falls and Bukal (Kilangin) Falls, with the highest scores, are the most suitable recommendations, indicating strong alignment with outdoor / adventure-seeking preferences.

Mount Banahaw–San Cristobal Protected Landscape and Mangulila Twin Falls are moderately recommended, showing relevance but slightly fewer contextual matches. Locations like APA Wildlife Rescue and Conservation Park, Malinao Resort, APA Farms & Leisure, and Resort River Front received lower scores, suggesting they are less strongly aligned with the current context. Overall, the scores demonstrate the system's ability to prioritize POIs effectively, highlighting the most relevant destinations while still offering alternatives.

Table 19. System Recommendations

ID	Name	Visitors	Score
1	Sprite Falls	High	1.500
2	Bukal (Kilangin) Falls	Very High	1.500
3	Mount Banahaw–San Cristobal Protected Landscape	Very High	1.300
4	Mangulila Twin Falls	High	1.100
5	APA Wildlife Rescue and Conservation Park	Moderate	0.800
6	Malinao Resort	Moderate	0.800
7	APA Farms & Leisure	Moderate	0.800
8	Resort River Front	Moderate	0.800

Testing and Evaluation of the System using ISO 25010 and TAM

This section presents the testing and evaluation results of the Majayjay Tourism Circuit Web Platform based on two recognized assessment frameworks: ISO 25010 and the Technology Acceptance Model (TAM). ISO 25010 was used to evaluate the technical quality of the system, including functionality, reliability, usability, efficiency, maintainability, and security.

Meanwhile, TAM was applied to measure user acceptance by assessing perceived usefulness, perceived ease of use, and overall willingness to adopt the platform. The combined results from both evaluations provide a comprehensive understanding of the system's performance and its acceptance by the intended users.

ISO 25010 Results

Table 20 shows that the system is able to perform all its core functions correctly and reliably. Users were satisfied with the accuracy and completeness of the information provided about tourist spots and accommodations. The Machine Learning recommendation system also performed well by giving relevant and useful suggestions.

The mean scores range from 4.33 to 4.67, all of which are interpreted as *Very Satisfied*. These results indicate that users are highly satisfied with the system's functionality. Based on the ISO 25010 standard, the system demonstrates high functional suitability, as it meets the required functions and delivers reliable and appropriate outputs. Overall, the system shows good performance and dependability.

Table 20. Functional Sustainability

Item	Mean	Standard Deviation	Interpretation
The system performs all its intended core functions completely and reliably.	4.67	0.22	Very Satisfied
The information provided about tourist spots and accommodations is accurate and comprehensive.	4.33	0.22	Very Satisfied
The Machine Learning recommendation system provides relevant and useful suggestions.	4.67	0.22	Very Satisfied
All functional requirements are fully met by the system.	4.67	0.22	Very Satisfied

Table 21 shows that the system responds quickly to user interactions, loads pages and processes data efficiently, handles multiple tasks simultaneously, and remains responsive even under heavy usage. The mean scores range from 4.33 to 5, which indicates that users are very satisfied with the system's performance. According to ISO 25010, this means the system demonstrates performance efficiency, as it uses resources effectively and provides fast, reliable responses under various conditions.

Table 21. Performance Efficiency

Item	Mean	Standard Deviation	Interpretation
The system responds quickly to user interactions.	5	0	Very Satisfied
Page loading and processing times are fast and efficient.	4.67	0.22	Very Satisfied
The system handles multiple tasks simultaneously without performance issues.	4.33	0.23	Very Satisfied
The platform remains responsive even under heavy usage.	4.33	0.23	Very Satisfied

Table 22 shows that the system works well on different devices and operating systems, functions consistently across web browsers, and integrates smoothly with other platforms. With mean scores from 4.33 to 4.67, users are very satisfied. Based on ISO 25010, the system demonstrates high compatibility, operating effectively in different environments.

Table 22. Compatibility

Item	Mean	Standard Deviation	Interpretation
The system works well on different devices (desktop, laptop, tablet, smartphone).	4.33	0.45	Very Satisfied
The platform functions consistently across different web browsers.	4.67	0.22	Very Satisfied
The system integrates seamlessly with other required platforms.	4.67	0.22	Very Satisfied
All features work properly on different operating systems.	4.33	0.23	Very Satisfied

Table 23 shows that users find the system easy to use and understand. The interface is intuitive and easy to navigate, allowing users to move through the platform without confusion. Learning how to use the system requires very little time and effort, even for first-time users.

The interactive map is user-friendly and helps users easily explore tourist locations. Common tasks such as searching for destinations, viewing details, and booking are straightforward and easy to complete. In addition, the system provides clear instructions and helpful error messages, which guide users when mistakes occur. Overall, the results indicate that the system has high usability, making it efficient and comfortable for users to achieve their goals.

Table 23. Usability

Item	Mean	Standard Deviation	Interpretation
The platform interface is intuitive and easy to navigate.	4.67	0.22	Very Satisfied
Learning to use the system requires minimal time and effort.	5.00	0.00	Very Satisfied
The interactive map is user-friendly and easy to understand.	4.67	0.22	Very Satisfied
Completing tasks (searching, booking, etc.) is straightforward.	4.67	0.22	Very Satisfied
Error messages and instructions are clear and helpful.	4.67	0.22	Very Satisfied

Table 24 indicate that the system operates stably without crashes, key features work consistently, performance is maintained during extended use, and data is saved correctly without loss. With mean scores ranging from 4.67 to 5.00, users are very satisfied.

Table 24. Reliability

Item	Mean	Standard Deviation	Interpretation
The system operates stably without unexpected crashes.	4.67	0.22	Very Satisfied
Key features work reliably without errors.	4.67	0.22	Very Satisfied
The platform maintains performance during extended use.	4.67	0.22	Very Satisfied
Data is saved correctly and not lost during operations.	5.00	0.00	Very Satisfied

Table 25 indicate that users feel confident their personal information is protected, login credentials and data are securely stored, unauthorized access is prevented, and sensitive transactions are handled safely. With mean scores of 4.67, users are very satisfied. According to ISO 25010,

Table 25. Security

Item	Mean	Standard Deviation	Interpretation
I feel confident that my personal information is protected.	4.67	0.22	Very Satisfied
Login credentials and user data are securely stored.	4.67	0.22	Very Satisfied
The system has adequate protection against unauthorized access.	4.67	0.22	Very Satisfied
Sensitive transactions are handled securely.	4.67	0.22	Very Satisfied

Table 26 shows that the system can be updated and modified easily, error handling and debugging are simple, the code structure is organized for future improvements, and adding new features is manageable. With mean scores of 4.67, users are very satisfied. According to ISO 25010, the system demonstrates strong maintainability, allowing for smooth updates, enhancements, and long-term maintenance.

Table 26. Maintainability

Item	Mean	Standard Deviation	Interpretation
The system appears easy to update and modify.	4.67	0.22	Very Satisfied
Error handling and debugging seem straightforward.	4.67	0.22	Very Satisfied
The codebase appears well-organized for future enhancements.	4.67	0.22	Very Satisfied
Adding new features seems manageable.	4.67	0.22	Very Satisfied

Table 27 shows that the system is easy to install, transfer, and adapt to different environments. The perfect mean score of 5.00 indicates very high user satisfaction. Based on ISO 25010, the system demonstrates excellent portability and can be deployed across different platforms with ease.

Table 27. Portability

Item	Mean	Standard Deviation	Interpretation
The system can be easily transferred to different environments.	5.00	0.00	Very Satisfied
Installation and setup appear straightforward	5.00	0.00	Very Satisfied
The platform adapts well to different technical configurations.	5.00	0.00	Very Satisfied
Migration to different servers seems feasible.	5.00	0.00	Very Satisfied

Table 28 indicates that users are satisfied with the quality of the tourism platform and are woulding to recommend it to others. The mean scores range from 3.67 to 5.00, showing very high satisfaction. According to ISO 25010, this reflects a positive overall perception of the system's quality and user experience.

Table 28. Overall Assessment

Item	Mean	Standard Deviation	Interpretation
Overall, I am satisfied with the quality of the tourism platform.	3.67	0.22	Satisfied
The system meets my expectations for a web-based tourism platform.	5.00	0.00	Very Satisfied
I would recommend this platform to other users.	5.00	0.00	Very Satisfied

Technology Acceptance Model (TAM) Results

Table 29 shows that users have a very positive perception of the platform, with mean scores ranging from 4.16 to 4.41, indicating strong to very strong agreement on its usefulness. Users rated the platform highest for helping them discover hidden tourist spots that match their interests, showing that the recommendation features are effective. which helps users make better travel decisions, and for supporting itinerary management..

The platform is also viewed as improving the overall efficiency of trip planning by saving time and effort. Although the score for increasing on-site enjoyment is slightly lower, it still reflects a positive impact, suggesting that the system is more strongly valued as a planning and discovery tool rather than for influencing experiences during the trip itself. and overall results confirm that the platform is effective and valuable for tourism planning and discovery.

Table 29. Result of Perceived usefulness

ITEM	MEAN	SD	INTERPRETATION
The platform improves the efficiency of planning trips to Majayjay, Laguna.	4.25	0.88	Strongly Agree
The platform provides accurate and comprehensive information that enhances decision-making for tourism activities.	4.31	0.76	Strongly Agree
Using this platform increases my enjoyment and satisfaction when visiting Majayjay.	4.16	0.98	Agree
The platform helps discover hidden tourist spots and activities that match my interests.	4.41	0.85	Strongly Agree
I find the platform valuable in managing my travel itinerary effectively.	4.26	0.78	Strongly Agree

Table 30 shows that users have a very positive perception of the platform, with mean scores ranging from 4.16 to 4.41, indicating strong to very strong agreement on its usefulness. The platform is especially effective in helping users discover hidden tourist spots that match their interests, which received the highest mean score. Users also rated the system highly for providing accurate information that supports better decision-making and for helping manage travel itineraries more efficiently.

The platform is also seen as improving the overall efficiency of trip planning by saving users time and effort. Although the score for increasing on-site enjoyment is slightly lower, it is still rated positively, suggesting that the platform is more effective as a planning and discovery tool than in directly influencing on-site experiences. and the high composite mean of approximately 4.28 confirms the platform's strong value and effectiveness in tourism planning and discovery.

Table 30. Perceived Ease of Use

ITEM	MEAN	SD	INTERPRETATION
The platform interface is user-friendly and easy to navigate.	4.08	0.96	Agree
I find it easy to learn how to use the platform and access its features.	4.2	0.91	Agree
Using the platform requires minimal effort or technical knowledge.	4.14	0.96	Agree
The interactive map makes it convenient to explore tourist spots in Majayjay.	4.39	0.85	Strongly Agree
I am comfortable with the overall design and functionality of the tourism platform.	4.23	0.93	Strongly Agree

The results indicate that users hold an exceptionally positive attitude towards the platform, with mean scores ranging from 4.16 to 4.46, reflecting strong agreement on its value and appeal. The most significant finding is users' strong wouldingness to advocate for the platform, as shown by the highest mean score (4.46) for recommending it to others. This is closely supported by a strong belief in its overall benefit for tourists (4.37) and the view that using it is an enjoyable and beneficial experience (4.30).

Users also feel positive about using it for their own trip planning (4.23). The slightly lower, though still high, score for fitting well with existing travel habits (4.16) suggests the platform integrates well but may represent a slight shift in routine for some. The consistently low standard deviations (0.83-0.91) show a strong consensus in these positive attitudes. The composite mean of 4.30 confirms that the platform is not only seen as useful and easy to use but is also genuinely well-liked and endorsed by its users.

Table 31. Attitude

ITEM	MEAN	SD	INTERPRETATION
I feel positive about using the platform for planning my trip to Majayjay.	4.23	0.87	Strongly Agree
The platform fits well with my travel planning habits.	4.16	0.91	Agree
I believe this technology can greatly benefit tourists visiting Majayjay.	4.37	0.83	Strongly Agree
Using this platform is an enjoyable and beneficial experience.	4.3	0.84	Strongly Agree
I would recommend this platform to other tourists.	4.46	0.83	Strongly Agree

The results demonstrate strong user behavioral intention to use the platform, with mean scores ranging from 4.03 to 4.35. Users show the highest level of intention towards adopting its specific features, as indicated by the strong agreement to use the machine learning recommendations (4.35) and to recommend the platform to others (4.33). The slightly lower, though still positive, scores for regular use (4.03), reliance on future visits (4.10), and commitment to integrating it into a routine (4.07) suggest that while users are very woulding to use and advocate for the platform, their commitment to making it a primary, habitual tool is strong but slightly more measured. The low standard deviations (0.77-0.92) reveal highly consistent feedback across all respondents. This overall positive behavioral intention, with a composite mean of approximately 4.18, confirms that the platform successfully motivates users to plan for its continued use and active promotion.

Table 32. Behavioural Intention

ITEM	MEAN	SD	INTERPRETATION
I intend to use this platform regularly for planning trips to Majayjay.	4.03	0.92	Agree
I am likely to rely on this platform to help plan my future visits to Majayjay.	4.1	0.88	Agree
I am woulding to use the machine learning recommendations feature for discovering new places.	4.35	0.77	Strongly Agree
I am likely to recommend this platform to friends and family visiting Majayjay.	4.33	0.79	Strongly Agree
I am committed to using this platform as part of my travel planning routine.	4.07	0.90	Agree

CHAPTER 5

SUMMARY CONCLUSION AND RECOMMENDATIONS

Summary

The Majayjay Tourism Circuit platform was developed to provide tourists with an easy and efficient way to access information about tourist spots, accommodations, and activities in Majayjay. One of its key goals is to showcase Majayjay as a hidden gem, highlighting its natural beauty, cultural attractions, and local experiences to a wider audience. The platform was developed using the Agile Scrum methodology, which allowed the researchers to work iteratively, gather continuous feedback, and implement improvements during each sprint. This approach ensured that the system was built in a flexible, user-centered manner, responding to both technical requirements and user needs throughout the development process.

The platform is composed of five main modules, each serving a specific purpose: the Account Module enables users to register, log in and out, and manage their profiles securely; the Tourist Listing Module provides accurate and comprehensive information about tourist spots and allows users to search and filter listings efficiently; the Accommodation Management Module allows administrators to manage accommodations and ensures users can view complete and updated details; the Recommendation Module offers personalized suggestions based on user preferences, helping tourists plan their trips effectively; and the Admin Management Module allows administrators to manage users, monitor listings, and oversee system performance, ensuring smooth platform operations.

To evaluate the system, the researchers conducted system testing for all modules and gathered user feedback using the Technology Acceptance Model (TAM). System testing confirmed that each module performed as expected, with all functionalities operating correctly and efficiently. Users were able to complete key tasks, such as registering accounts, searching for tourist spots, managing accommodations, receiving recommendations, and performing administrative operations without encountering errors. These results demonstrate that the platform is reliable, functional, and well-integrated across all its modules.

Ethical considerations were also addressed during the study. Users' personal information was handled securely, with login credentials and profile data protected to prevent unauthorized access. Participation in the evaluation was voluntary, and respondents were informed about the purpose of the study. Data collected from users was kept confidential and used solely for evaluating the system's usability, functionality, and effectiveness. These measures ensured compliance with ethical research standards and promoted user trust in the platform.

Compatibility testing was also assessed through user feedback based on the Technology Acceptance Model (TAM). Users accessed the system on different devices such as desktops, laptops, tablets, and smartphones, and on various browsers including Google Chrome, Mozilla Firefox, and Microsoft Edge. Feedback indicated that the system performed consistently across devices and browsers, with features working correctly and interfaces displaying

properly. This shows that the platform is flexible and portable, ensuring a good user experience regardless of technical environment.

System testing was conducted to ensure that all modules functioned correctly according to their intended purpose. Each module was evaluated for expected output versus actual output. Results showed that the Account Module allowed users to register and manage accounts without issues, the Tourist Listing Module displayed accurate information and functional search/filter options, the Accommodation Management Module allowed proper CRUD operations, the Recommendation Module provided relevant and personalized suggestions, and the Admin Management Module enabled smooth administrative tasks. All modules passed testing, confirming the system is reliable, fully functional, and integrated across its components.

Conclusion

The development and evaluation of the Majayjay Tourism Circuit platform achieved its objectives of providing tourists with easy access to information on tourist spots, accommodations, and activities in Majayjay, while also showcasing the town as a hidden gem. Developed using the Agile Scrum methodology, the platform ensures iterative improvements, user-centered design, and effective integration of all its modules.

The platform’s five main modules, Account, Tourist Listing, Accommodation Management, Recommendation, and Admin Management, were tested and confirmed to function correctly, reliably, and efficiently. System testing showed that all features operated as intended, supporting user tasks and administrative operations without errors. Compatibility testing, evaluated

through user feedback, demonstrated that the platform performs consistently across different devices and browsers, providing a seamless user experience.

Through the Technology Acceptance Model (TAM) evaluation, users reported satisfactory answer with the system's usability, usefulness, security, and portability. They found it easy to navigate, appreciated the interactive map and personalized recommendations, and expressed confidence in the platform's overall functionality.

Overall, the study concludes that the Majayjay Tourism Circuit platform is technically reliable, user-friendly, and effective in promoting tourism in Majayjay. The combination of thorough system testing, positive user acceptance, and adherence to ethical standards confirms that the platform is a valuable tool for both tourists and administrators, contributing to enhanced tourism experiences and the promotion of Majayjay as a hidden gem.

Recommendation

To ensure the Majayjay Tourism Circuit platform continues to meet the needs of its users, this section provides a set of recommendations. These suggestions are based on system performance, user experience, and feedback, and are intended to improve the platform's usefulness, accessibility, and overall impact. By following these recommendations, the system can better support tourists, local businesses, and government stakeholders while addressing current limitations and guiding future enhancements.

1. It is recommended that tourists use the platform to discover hidden attractions, plan their trips efficiently, and make informed decisions using the provided information, personalized suggestions, and community feedback.

2. The platform should encourage collaboration with local businesses and the LGU to include real-time deals, seasonal events, and special packages, which can attract more visitors and support the local economy.

3. Regular updates, verification of tourist and accommodation information, and improvements to the user interface—such as tooltips, tutorials, or onboarding guides—are suggested to maintain reliability, usability, and credibility for both tourists and local authorities.

4. Improving the user interface with tooltips, tutorials, or onboarding guides is recommended to help new users navigate the platform more easily and enhance overall usability.

5. Strengthening data security and privacy measures is suggested to protect user information and increase trust in the system.

6. Future research can explore advanced recommendation techniques, study user adoption and satisfaction, and gather feedback from tourists to further improve personalization, system performance, and overall user experience.

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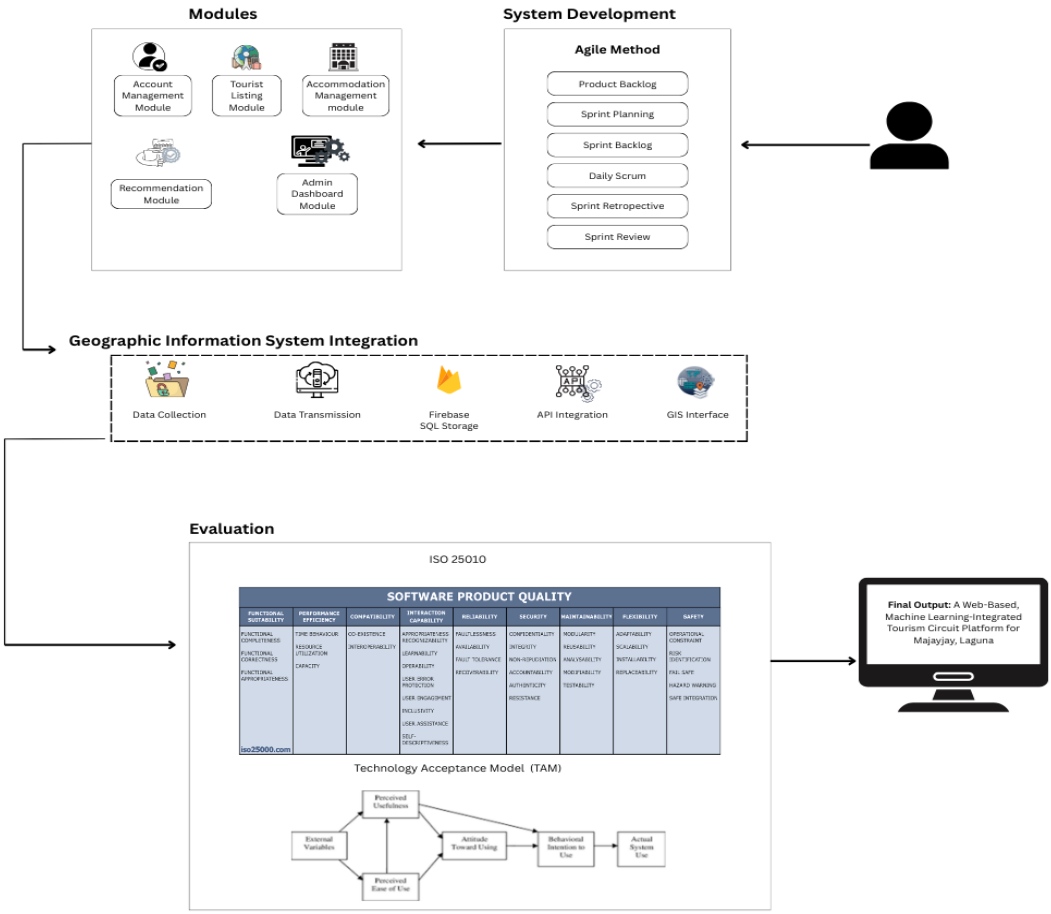
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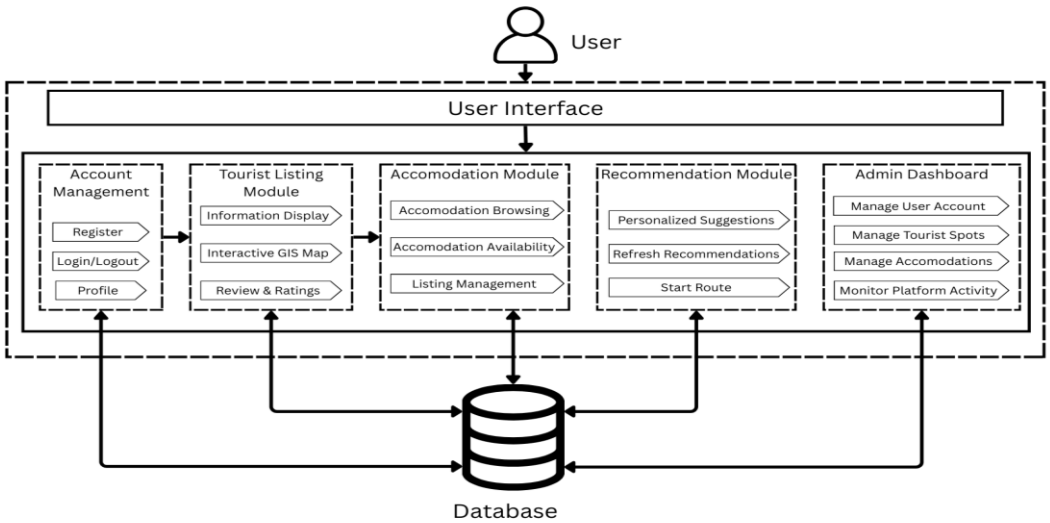
APPENDIX A.

Figures of System Design, Algorithms, and User Interfaces

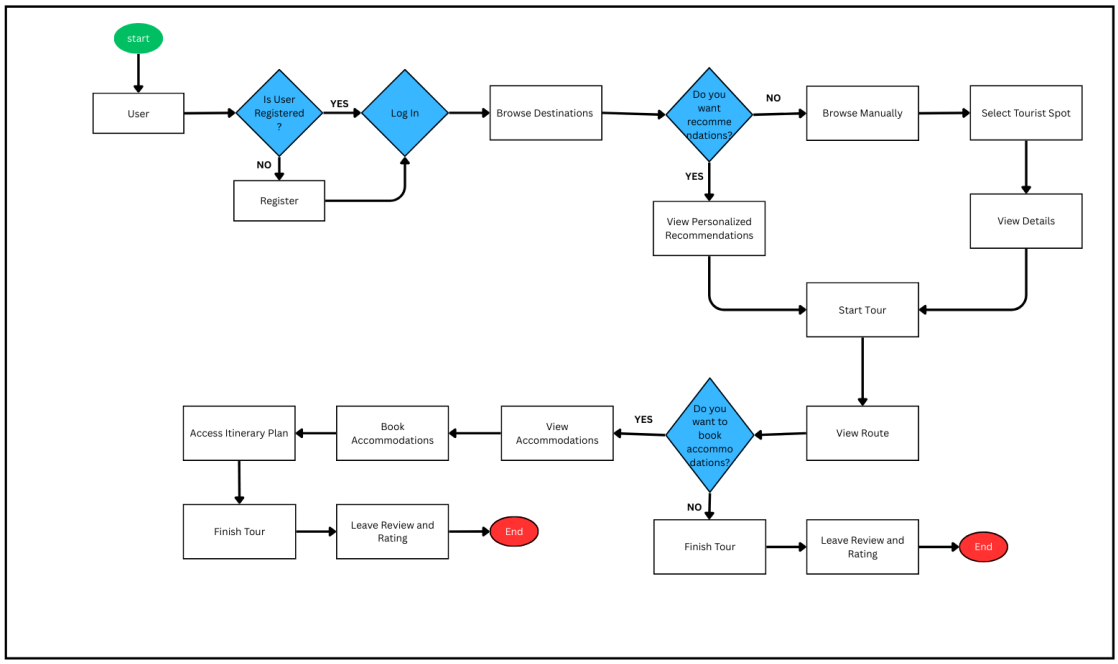
Conceptual Framework



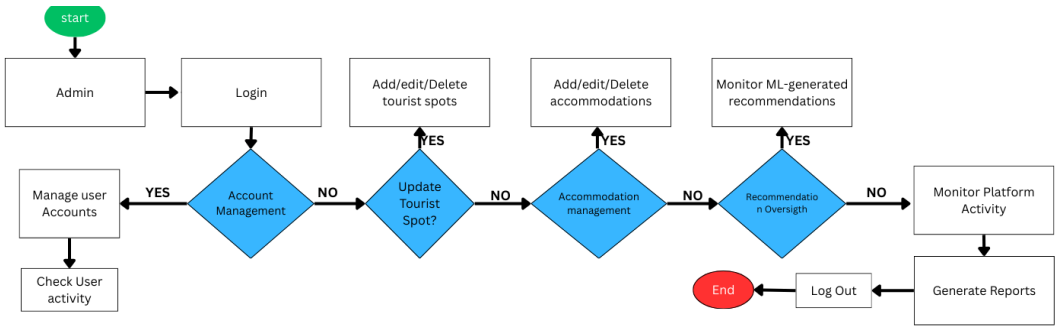
System Architecture



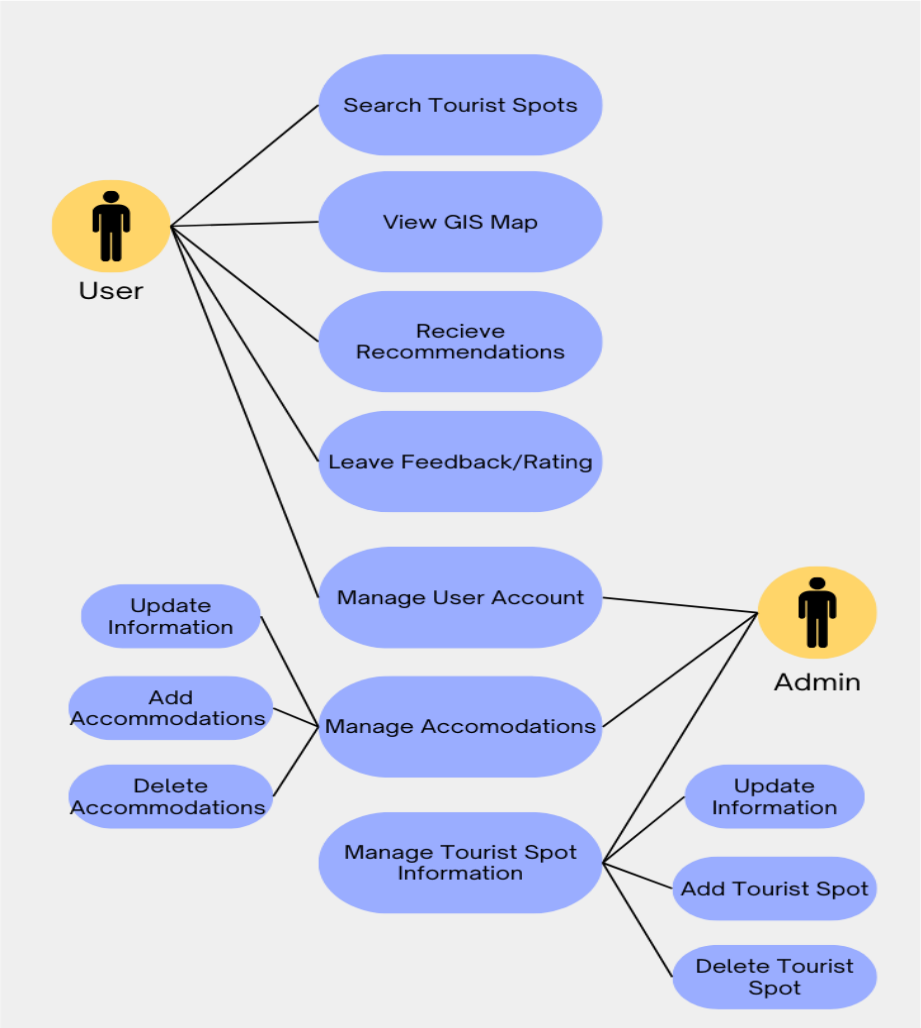
Tourist Flowchart



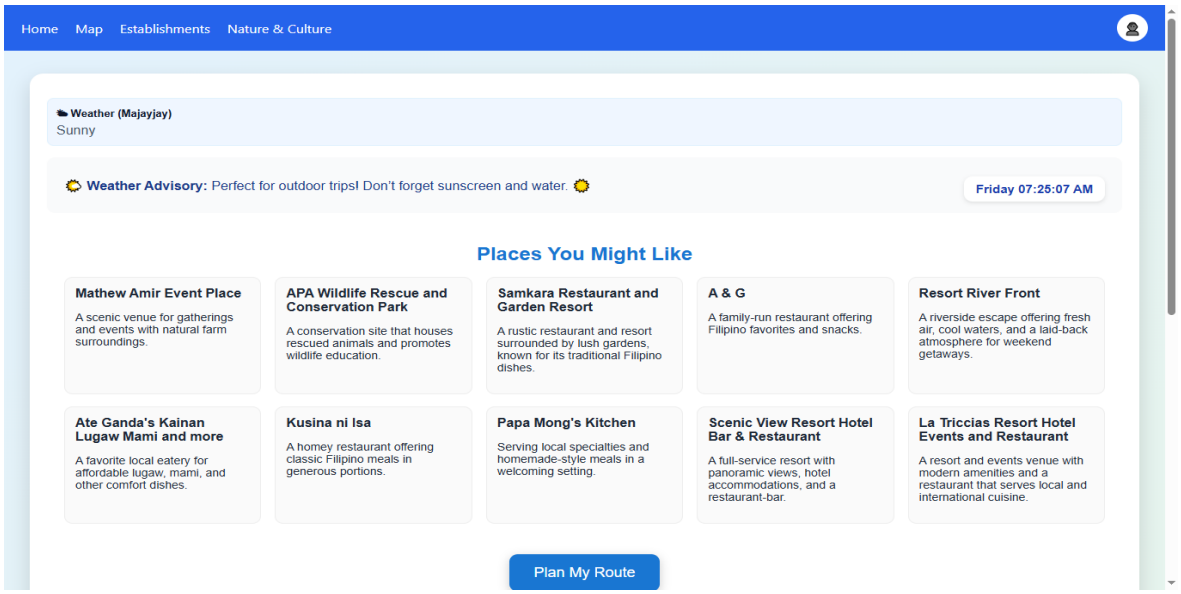
Admin Flowchart

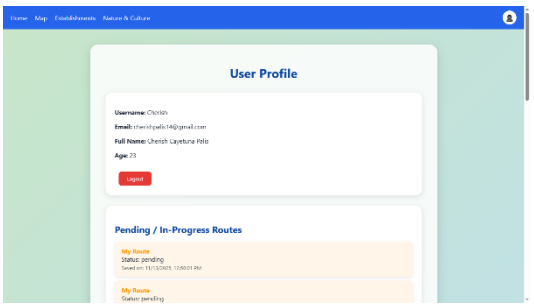


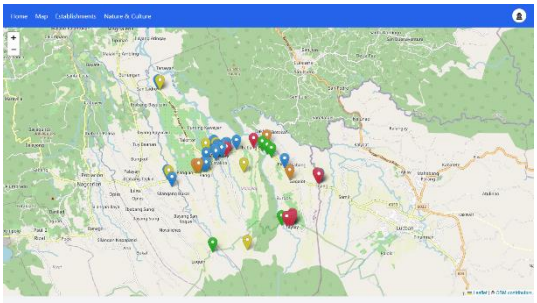
Use Case Diagram

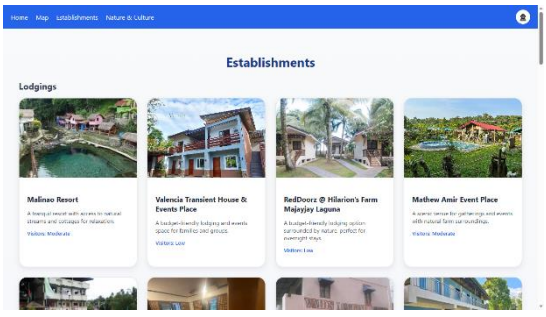


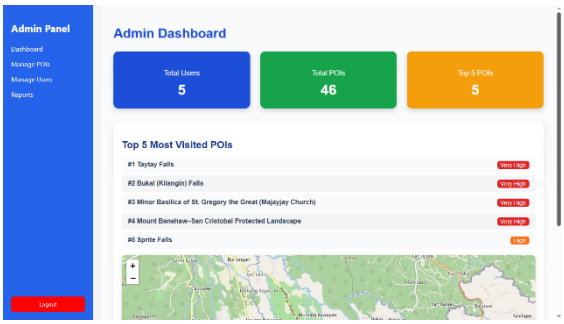
User Interface



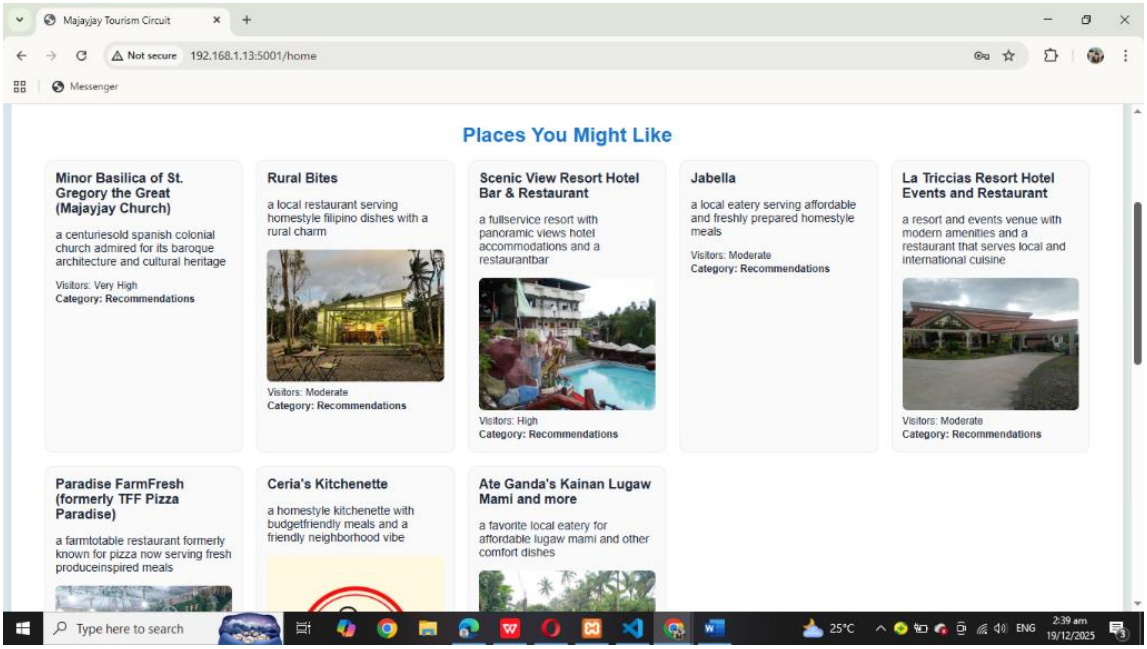


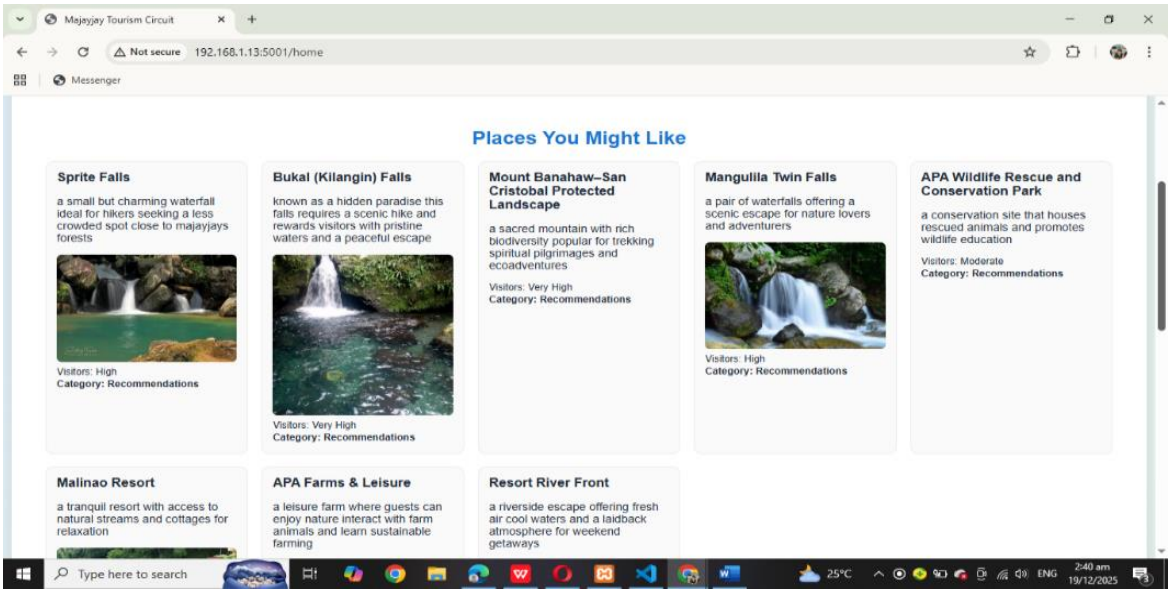






Smart Recommedation Interface





APPENDIX B.

Survey Questionnaires

Techonology Acceptance Questionnaire

TECHNOLOGY ACCEPTANCE MODEL (TAM) QUESTIONNAIRE
For Web-Based Tourism Platform of Majayjay, Laguna

Dear Respondent,

We are students of the Laguna State Polytechnic University -- Los Baños Campus under the College of Computer Studies conducting development on the project entitled **"IMPLEMENTING TOURISM CIRCUIT CONCEPT WITH MACHINE LEARNING ON A WEB-BASED TOURISM PLATFORM FOR MAJAYJAY, LAGUNA"**. We would like to seek your opinion on the acceptability of the said project by scoring the statements below by checking the appropriate numbers, wherein:

1 = Strongly Disagree; 2 = Disagree; 3 = Neutral; 4 = Agree; 5 = Strongly Agree

Rest assured that any information that you provide would be treated with utmost confidentiality (RA 10173. Data Privacy Act) and would be used for research purposes only. Thank you for your cooperation.

Researchers: Rel Sallum M. Aragon, John Patrick M. Tandang, Cherish C. Palis

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PERCEIVED USEFULNESS

No.	Statement	1 Strongly Disagree	2 Disagreeu	3 Neutral	4 Agree	5 Strongly Agree
1.	The platform improves the efficiency of planning trips to Majayjay, Laguna. *Pinapahusay ng platform ang kahusayan sa pagpaplano ng mga biyahe patungong Majayjay, Laguna.*					
2.	The platform provides accurate and comprehensive information that enhances decision-making for tourism activities. *Nagbibigay ang platform ng tumpak at komprehensibong impormasyon na nagpapabuti sa paggawa ng desisyon para sa mga gawaing panturismo.*					
3.	Using this platform increases my enjoyment and satisfaction when visiting Majayjay. *Ang paggamit ng platform na ito ay nagpapataas ng aking kasiyahan at kagalakan kapag bumibisita sa Majayjay.*					
4.	The platform helps discover hidden tourist spots and activities that match my interests. *Nakakatulong ang platform na matuklasan ang mga nakatagong tourist spot at aktibidad na akma sa aking mga interes.*					

5.	I find the platform valuable in managing my travel itinerary effectively. *Mahalaga sa akin ang platform na ito sa mahusay na pamamahala ng aking itineraryo ng paglalakbay.*					

PERCEIVED EASE OF USE

No.	Statement	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
1.	The platform interface is user-friendly and easy to navigate. *Ang interface ng platform ay madaling gamitin at madaling i-navigate.*					
2.	I find it easy to learn how to use the platform and access its features. *Madali kong natutunan kung paano gamitin ang platform at ma-access ang mga feature nito.*					
3.	Using the platform requires minimal effort or technical knowledge. *Ang paggamit ng platform ay nangangailangan ng kaunting pagsisikap o teknikal na kaalaman.*					
4.	The interactive map makes it convenient to explore tourist spots in Majayjay. *Ginagawa ng interactive map na maginhawa ang pagtuklas sa mga tourist spot sa Majayjay.*					
5.	I am comfortable with the overall design and functionality of the tourism platform. *Komportable ako sa kabuuang disenyo at functionality ng tourism platform.*					

ATTITUDE (TOWARDS USING THE SYSTEM)

No.	Statement	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
1.	I feel positive about using the platform for planning my trip to Majayjay. *Pakiramdam ko ay positibo ang paggamit ng platform para sa pagpaplano ng aking biyahe sa Majayjay.*					

2.	The platform fits well with my travel planning habits. *Ang platform ay angkop sa aking mga nakagawiang paraan ng pagpaplano ng paglalakbay.*					
3.	I believe this technology can greatly benefit tourists visiting Majayjay. *Naniniwala ako na ang teknolohiyang ito ay malaking maitutulong sa mga turistang bumibisita sa Majayjay.*					
4.	Using this platform is an enjoyable and beneficial experience. *Ang paggamit ng platform na ito ay isang kasiya-siya at kapaki-pakinabang na karanasan.*					
5.	I would recommend this platform to other tourists. *Irekomenda ko ang platform na ito sa iba pang mga turista.*					

BEHAVIORAL INTENTION (TO USE THE SYSTEM)

No.	Statement	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
1.	I intend to use this platform regularly for planning trips to Majayjay. *Balak kong gamitin ang platform na ito nang regular para sa pagpaplano ng mga biyahe sa Majayjay.*					
2.	I am likely to rely on this platform to help plan my future visits to Majayjay. *Malamang na aasa ako sa platform na ito upang makatulong sa pagpaplano ng aking mga pagbisita sa Majayjay sa hinaharap.*					
3.	I am woulding to use the machine learning recommendations feature for discovering new places. *Handa akong gamitin ang machine learning recommendation feature para matuklasan ang mga bagong lugar.*					
4.	I am likely to recommend this platform to friends and family visiting Majayjay. *Malamang na irekomenda ko ang platform na ito sa mga kaibigan at pamilyang bibisita sa Majayjay.*					

I am committed to using this platform as part of my travel planning routine.

5. *Nais kong gamitin ang platform na ito bilang bahagi ng aking nakagawiang paraan ng pagpaplano ng paglalakbay.*

RESPONDENT INFORMATION

Name of Respondent: (optional)_____

Signature: _____

Date: _____

Role:
Tourist
Local Business Owner
LGU Representative
IT Expert
Other: _____

ISO Questionnaire

SYSTEM EVALUATION QUESTIONNAIRE

PROJECT TITLE: **[IMPLEMENTING TOURISM CIRCUIT CONCEPT WITH MACHINE LEARNING ON A WEB-BASED TOURISM PLATFORM FOR MAJAYJAY, LAGUNA]**

THE RESEARCHERS: [**Rel Sallum M. Aragon, John Patrick M. Tandang, Cherish C. Palis**]

Good Day! We are 4th year students, currently pursuing a Bachelor of Science in Information Technology specializing in System Mangement Program. As part of our academic endeavors, we have developed a web-based system adhering to the ISO 25010 standard.

To assess the effectiveness and user satisfaction of our system, we have devised a set of questions. Your responses, rated on a scale from 1 to 5, would help us evaluate various aspects of the system, from usability, functionality, security, and overall user experience.

(Based on ISO 25010 Software Quality Standard)

Instructions: Thank you for participating in the evaluation of our web-based tourism platform. Please rate the following aspects based on your experience with the system. Check your answer using the scale provided below.

Rating Scale:

1 = Poor 2= Needs Improvement 3 = Fair 4 = Good 5 = Excellent
RUBRIC FOR 5-Point Likert Scale to evaluate product/system quality using ISO 25010

FUNCTIONAL SUITABILITY

No.	Statement	1 Poor	2 Needs Improvement	3 Fair	4 Good	5 Excellent
1.	The system performs all its intended core functions completely and reliably.					
2.	The information provided about tourist spots and accommodations is accurate and comprehensive.					
3.	The Machine Learning recommendation system provides relevant and useful suggestions.					
4.	All functional requirements are fully met by the system.					

PERFORMANCE EFFICIENCY

No.	Statement	1 Poor	2 Needs Improvement	3 Fair	4 Good	5 Excellent
1.	The system responds quickly to user interactions.					
2.	Page loading and processing times are fast and efficient.					
3.	The system handles multiple tasks simultaneously without performance issues.					
4.	The platform remains responsive even under heavy usage.					

COMPATIBILITY

No.	Statement	1 Poor	2 Needs Improvement	3 Fair	4 Good	5 Excellent
1.	The system works well on different devices (desktop, laptop, tablet, smartphone).					
2.	The platform functions consistently across different web browsers.					
3.	The system integrates seamlessly with other required platforms.					
4.	All features work properly on different operating systems.					

USABILITY

No.	Statement	1 Poor	2 Needs Improvement	3 Fair	4 Good	5 Excellent
1.	The platform interface is intuitive and easy to navigate.					
2.	Learning to use the system requires minimal time and effort.					
3.	The interactive map is user-friendly and easy to understand.					
4.	Completing tasks (searching, booking, etc.) is straightforward.					
5.	Error messages and instructions are clear and helpful.					

RELIABILITY

No.	Statement	1 Poor	2 Needs Improvement	3 Fair	4 Good	5 Excellent
1.	The system operates stably without unexpected crashes.					
2.	Key features work reliably without errors.					
3.	The platform maintains performance during extended use.					
4.	Data is saved correctly and not lost during operations.					

SECURITY

No.	Statement	1 Poor	2 Needs Improvement	3 Fair	4 Good	5 Excellent
1.	I feel confident that my personal information is protected.					
2.	Login credentials and user data are securely stored.					

3.	The system has adequate protection against unauthorized access.					
4.	Sensitive transactions are handled securely.					

MAINTAINABILITY

(Note: These statements are primarily for IT Expert respondents)

No.	Statement	1 Poor	2 Needs Improvement	3 Fair	4 Good	5 Excellent
1.	The system appears easy to update and modify.					
2.	Error handling and debugging seem straightforward.					
3.	The codebase appears well-organized for future enhancements.					
4.	Adding new features seems manageable.					

PORTABILITY

(Note: These statements are primarily for IT Expert respondents)

No.	Statement	1 Poor	2 Needs Improvement	3 Fair	4 Good	5 Excellent
1.	The system can be easily transferred to different environments.					
2.	Installation and setup appear straightforward.					
3.	The platform adapts well to different technical configurations.					
4.	Migration to different servers seems feasible.					

OVERALL ASSESSMENT

No.	Statement	1 Poor	2 Needs Improvement	3 Fair	4 Good	5 Excellent
1.	Overall, I am satisfied with the quality of the tourism platform.					

2. The system meets my expectations for a web-based tourism platform.

3. I would recommend this platform to other users.

ADDITIONAL COMMENTS

What are the key strengths of the system?

What are your main recommendations for improvement?

RESPONDENT INFORMATION

Name of Respondent:(optional): _____

Signature: _____

Date: _____

Role:

Tourist
Local Business Owner
LGU Representative
IT Expert
Other: _____

APPENDIX C.

Relevant Source Code

```
import pandas as pd
import numpy as np
import matplotlib
matplotlib.use('Agg')  # Set backend to Agg for server use
import matplotlib.pyplot as plt
import requests
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.preprocessing import StandardScaler
from datetime import datetime
import re
import io
import base64
from flask_cors import CORS
from flask import Flask, jsonify, request

app = Flask(__name__)
CORS(app)

class POIRecommendationSystem:
    def __init__(self):
        self.df = None
        self.feature_matrix = None
        self.scaler = StandardScaler()
        self.vectorizer = TfidfVectorizer(max_features=30, stop_words='english',
ngram_range=(1, 1))
        self.temporal_context = {}
        self.weather_condition = "unknown"

        # Knowledge base
        self.seasonal_patterns = {
            'summer': {'mar', 'apr', 'may', 'jun'},
            'rainy': {'jul', 'aug', 'sep', 'oct'},
            'cool_dry': {'nov', 'dec', 'jan', 'feb'}
        }

        self.time_preferences = {
            'morning': {'hiking', 'adventure', 'nature', 'religious'},
            'afternoon': {'waterfall', 'swimming', 'coffee', 'restaurant'},
            'evening': {'restaurant', 'coffee', 'resort', 'hotel'},
            'weekend': {'adventure', 'waterfall', 'nature', 'swimming'},
            'weekday': {'coffee', 'restaurant', 'religious', 'historical'}
        }

        self.weather_recommendations = {
            'sunny': {'waterfall', 'swimming', 'nature', 'adventure', 'hiking'},
            'rainy': {'coffee', 'restaurant', 'religious', 'historical', 'indoor'},
            'cloudy': {'nature', 'farm', 'historical', 'coffee'},
            'stormy': {'restaurant', 'coffee', 'hotel', 'indoor'}
        }

    def get_pois_from_api(self):
```

```
        try:
            response = requests.get('http://localhost:5001/api/pois')
            if response.status_code == 200:
                return response.json()
            else:
                print(f"API Error: {response.status_code}")
                return None
        except requests.exceptions.RequestException as e:
            print(f"Request failed: {e}")
            return None

    def load_and_clean_data(self):
        """Load and clean the POI data"""
        data = self.get_pois_from_api()
        if data is None:
            return False

        self.df = pd.DataFrame(data)

        # Handle missing values
        self.df['description'] = self.df['description'].fillna('')
        self.df['image'] = self.df['image'].fillna('no_image.jpg')
        self.df['tags'] = self.df['tags'].fillna('')

        # Clean text data
        self.df['description'] = self.df['description'].apply(
            lambda x: re.sub(r'[^\w\s]', '', str(x)).lower()
        )

        # Convert visitors to numerical scores
        visitor_mapping = {'Very High': 4, 'High': 3, 'Moderate': 2, 'Low': 1}
        self.df['visitors_score'] = self.df['visitors'].map(visitor_mapping)

        return True

    def get_detailed_categories(self, tags, name, description):
        categories = set()
        text = f"{tags} {name} {description}".lower()

        # Activity-based categories
        if any(word in text for word in ['waterfall', 'falls']):
            categories.add('water_activity')
            categories.add('outdoor')
        if any(word in text for word in ['swimming']):
            categories.add('water_activity')
            categories.add('recreational')
        if any(word in text for word in ['hiking', 'trekking', 'adventure']):
            categories.add('adventure')
            categories.add('outdoor')
        if any(word in text for word in ['nature', 'forest', 'mountain']):
            categories.add('nature')
            categories.add('outdoor')
```

```
            # Facility-based categories
        if any(word in text for word in ['restaurant', 'cafe', 'coffee', 'bistro', 'eatery']):
            categories.add('dining')
            categories.add('indoor')
        if any(word in text for word in ['resort', 'hotel', 'lodging', 'lodge', 'inn']):
            categories.add('accommodation')
        if any(word in text for word in ['farm']):
            categories.add('agricultural')
            categories.add('educational')
        if any(word in text for word in ['church', 'religious', 'basilica', 'parish']):
            categories.add('religious')
            categories.add('cultural')
        if any(word in text for word in ['historical', 'history', 'heritage', 'architecture']):
            categories.add('historical')
            categories.add('cultural')

        # Weather compatibility
        if any(word in text for word in ['indoor', 'restaurant', 'cafe', 'church', 'hotel', 'resort']):
            categories.add('rain_friendly')
        if any(word in text for word in ['waterfall', 'swimming', 'outdoor', 'nature']):
            categories.add('sunny_friendly')

        return list(categories)

    def create_enhanced_features(self):
        """Create enhanced categorization and features"""
        print("Creating enhanced categories...")

        # Apply categorization
        self.df['enhanced_categories'] = self.df.apply(
            lambda row: self.get_detailed_categories(row['tags'], row['name'], row['description']),
axis=1
        )

        # Create binary features for enhanced categories
        all_enhanced_categories = set()
        for cats in self.df['enhanced_categories']:
            if isinstance(cats, list):
                all_enhanced_categories.update(cats)

        print(f"Found {len(all_enhanced_categories)} unique enhanced categories")

        # Initialize all enhanced category columns with 0
        for cat in all_enhanced_categories:
            self.df[f'enhanced_{cat}'] = 0

        # Set to 1 where category exists
        for idx, categories in enumerate(self.df['enhanced_categories']):
            if isinstance(categories, list):
                for cat in categories:
                    if f'enhanced_{cat}' in self.df.columns:
                        self.df.at[idx, f'enhanced_{cat}'] = 1
```

```
    def create_tag_features(self):
        """Create tag-based features"""
        print("Creating tag features...")
        all_tags = set()
        for tags in self.df['tags']:
            if pd.notna(tags) and isinstance(tags, str):
                all_tags.update([tag.strip() for tag in tags.split(',')])

        print(f"Found {len(all_tags)} unique tags")

        # Create binary features for each tag
        for tag in all_tags:
            self.df[f'tag_{tag}'] = self.df['tags'].apply(
                lambda x: 1 if pd.notna(x) and isinstance(x, str) and tag in [t.strip() for t in x.split(',')]
else 0
            )

    def prepare_features(self):
        """Prepare the feature matrix for recommendation"""
        # Get enhanced category columns and tag columns
        enhanced_cols = [col for col in self.df.columns if col.startswith('enhanced_')]
        tag_cols = [col for col in self.df.columns if col.startswith('tag_')]

        # Ensure all columns are numeric
        for col in enhanced_cols + tag_cols + ['visitors_score']:
            self.df[col] = pd.to_numeric(self.df[col], errors='coerce').fillna(0)

        # Combine all features
        feature_columns = ['visitors_score'] + tag_cols + enhanced_cols

        # TF-IDF for descriptions
        print("Processing descriptions with TF-IDF...")
        description_features = self.vectorizer.fit_transform(self.df['description'])
        description_df = pd.DataFrame(
            description_features.toarray(),
            columns=[f'desc_{i}' for i in range(description_features.shape[1])],
            index=self.df.index
        )

        # Combine numerical and text features
        numerical_features = self.df[feature_columns].copy()
        numerical_features = pd.concat([numerical_features, description_df], axis=1)

        # Ensure all values are numeric and scale
        numerical_features = numerical_features.apply(pd.to_numeric, errors='coerce').fillna(0)
        self.feature_matrix = self.scaler.fit_transform(numerical_features)

        print(f"Final feature matrix shape: {self.feature_matrix.shape}")

    def map_weather_code(self, code):
        """Map WMO weather code to condition"""
```

```
    weather_mapping = {
        0: "clear",
        1: "mainly clear",
        2: "partly cloudy",
        3: "overcast",
        45: "foggy",
        48: "foggy",
        51: "drizzle",
        53: "drizzle",
        55: "drizzle",
        61: "rainy",
        63: "rainy",
        65: "rainy",
        80: "rainy",
        81: "rainy",
        82: "rainy",
        95: "stormy",
        96: "stormy",
        99: "stormy"
    }
    return weather_mapping.get(code, "unknown")

def get_weather_condition(self):
    """Get current weather condition"""
    try:
        # Get current location-based weather using Open-Meteo
        url = "https://api.open-meteo.com/v1/forecast"
        params = {
            'latitude': 14.5995,  # Manila coordinates
            'longitude': 120.9842,
            'current_weather': 'true'
        }

        response = requests.get(url, params=params, timeout=10)
        data = response.json()

        if 'current_weather' in data:
            weather_code = data['current_weather']['weathercode']
            temperature = data['current_weather']['temperature']

            # Convert weather code to condition
            weather_condition = self.map_weather_code(weather_code)

            print(f"Current temperature: {temperature}°C")
            print(f"Weather condition: {weather_condition}")
            return weather_condition
        else:
            return "unknown"

    except Exception as e:
        print(f" ✕ Error fetching weather data: {e}")
        return "unknown"
```

```
def update_temporal_context(self):
    """Update temporal context based on current datetime"""
    current_datetime = datetime.now()

    # Time of day
    hour = current_datetime.hour
    if 5 <= hour < 12:
        self.temporal_context['time_of_day'] = 'morning'
    elif 12 <= hour < 17:
        self.temporal_context['time_of_day'] = 'afternoon'
    elif 17 <= hour < 21:
        self.temporal_context['time_of_day'] = 'evening'
    else:
        self.temporal_context['time_of_day'] = 'night'

    # Day type
    if current_datetime.weekday() >= 5:
        self.temporal_context['day_type'] = 'weekend'
    else:
        self.temporal_context['day_type'] = 'weekday'

    # Season (Philippines context)
    month = current_datetime.strftime('%b').lower()
    if month in self.seasonal_patterns['summer']:
        self.temporal_context['season'] = 'summer'
    elif month in self.seasonal_patterns['rainy']:
        self.temporal_context['season'] = 'rainy'
    else:
        self.temporal_context['season'] = 'cool_dry'

    # Special occasions
    self.temporal_context['is_summer_peak'] = (month in ['apr', 'may'])
    self.temporal_context['is_christmas_season'] = (month in ['dec', 'jan'])

    # Get weather condition
    self.weather_condition = self.get_weather_condition()

def get_recommendations(self, n_recommendations=8):
    """Get recommendations based on current context"""
    if self.df is None or self.feature_matrix is None:
        return None

    # Calculate temporal boosts
    print("Calculating temporal boosts...")
    boost_scores = np.zeros(len(self.df))

    # Time of day boost
    time_prefs = self.time_preferences.get(self.temporal_context['time_of_day'], set())
    day_type_prefs = self.time_preferences.get(self.temporal_context['day_type'], set())

    # Weather boost
```

```
        weather_prefs = set()
        if self.weather_condition:
            weather_prefs = self.weather_recommendations.get(self.weather_condition, set())

        # Seasonal boost
        season_boost = {
            'summer': {'water_activity', 'outdoor', 'adventure', 'swimming'},
            'rainy': {'indoor', 'dining', 'cultural', 'rain_friendly'},
            'cool_dry': {'nature', 'adventure', 'outdoor', 'hiking'}
        }
        season_prefs = season_boost.get(self.temporal_context['season'], set())

        # Combine all preferences
        all_preferences =
time_prefs.union(day_type_prefs).union(weather_prefs).union(season_prefs)
        print(f"Applying boosts for: {all_preferences}")

        # Apply boosts
        for preference in all_preferences:
            enhanced_col = f'enhanced_{preference}'
            if enhanced_col in self.df.columns:
                boost_scores += self.df[enhanced_col].values * 0.3

            tag_col = f'tag_{preference}'
            if tag_col in self.df.columns:
                boost_scores += self.df[tag_col].values * 0.2

        # Create user profile and calculate similarity
        user_profile = np.zeros(self.feature_matrix.shape[1])
        base_similarity = cosine_similarity([user_profile], self.feature_matrix)[0]

        # Apply temporal boost
        final_scores = base_similarity + boost_scores

        print(f"Similarity scores calculated for {len(self.df)} places")

        # Get top recommendations
        top_indices = np.argsort(final_scores)[::-1][:n_recommendations]

        recommendations = self.df.iloc[top_indices][['id', 'name', 'tags', 'visitors', 'description',
'image']].copy()
        recommendations['recommendation_score'] = final_scores[top_indices]

        # Add context reasons
        context_reasons = []
        for idx in top_indices:
            place = self.df.iloc[idx]
            reasons = []

            # Time-based reasons
            time_of_day = self.temporal_context.get('time_of_day', '')
            if time_of_day in ['morning', 'afternoon']:
```

```
                if self.df.at[idx, 'enhanced_adventure'] == 1 or self.df.at[idx, 'enhanced_nature'] ==
1:
                    reasons.append("Great for daytime adventures")

            if time_of_day == 'evening':
                if self.df.at[idx, 'enhanced_dining'] == 1:
                    reasons.append("Perfect for evening dining")

            # Weather-based reasons
            if self.weather_condition == 'sunny' and self.df.at[idx, 'enhanced_water_activity'] == 1:
                reasons.append("Ideal for sunny days")
            elif self.weather_condition == 'rainy' and self.df.at[idx, 'enhanced_rain_friendly'] == 1:
                reasons.append("Rain-friendly location")

            # Seasonal reasons
            season = self.temporal_context.get('season', '')
            if season == 'summer' and self.df.at[idx, 'enhanced_water_activity'] == 1:
                reasons.append("Perfect summer destination")
            elif season == 'rainy' and self.df.at[idx, 'enhanced_indoor'] == 1:
                reasons.append("Great for rainy season")

            context_reasons.append("; ".join(reasons) if reasons else "Popular destination")

        recommendations['context_reason'] = context_reasons

        return recommendations

    def initialize_system(self):
        """Initialize the recommendation system"""
        print("Initializing POI Recommendation System...")

        if not self.load_and_clean_data():
            return False

        self.create_enhanced_features()
        self.create_tag_features()
        self.prepare_features()
        self.update_temporal_context()

        print("POI Recommendation System initialized successfully!")
        return True

# Initialize the recommendation system
recommendation_system = POIRecommendationSystem()

# Flask Routes
@app.route('/api/recommendations', methods=['GET'])
def get_recommendations():
    """Get POI recommendations based on current context"""
    try:
        n_recommendations = request.args.get('limit', default=8, type=int)
```

```
        # Update temporal context for fresh recommendations
        recommendation_system.update_temporal_context()

        recommendations =
recommendation_system.get_recommendations(n_recommendations)

        if recommendations is None:
            return jsonify({'error': 'Recommendation system not initialized'}), 500

        # Convert recommendations to dictionary
        result = {
            'context': {
                'temporal': recommendation_system.temporal_context,
                'weather': recommendation_system.weather_condition
            },
            'recommendations': recommendations.to_dict('records')
        }

        return jsonify(result)

    except Exception as e:
        return jsonify({'error': str(e)}), 500

@app.route('/api/system/status', methods=['GET'])
def get_system_status():
    """Get system status"""
    status = {
        'initialized': recommendation_system.df is not None,
        'data_points': len(recommendation_system.df) if recommendation_system.df else 0,
        'feature_dimensions': recommendation_system.feature_matrix.shape if
recommendation_system.feature_matrix else None,
        'temporal_context': recommendation_system.temporal_context,
        'weather_condition': recommendation_system.weather_condition
    }
    return jsonify(status)

@app.route('/api/system/initialize', methods=['POST'])
def initialize_system():
    """Initialize or reinitialize the recommendation system"""
    try:
        success = recommendation_system.initialize_system()
        if success:
            return jsonify({'message': 'System initialized successfully'})
        else:
            return jsonify({'error': 'Failed to initialize system'}), 500
    except Exception as e:
        return jsonify({'error': str(e)}), 500

@app.route('/api/analysis/visualization', methods=['GET'])
def get_analysis_visualization():
    """Generate analysis visualizations"""
    try:
```

```
        if recommendation_system.df is None:
            return jsonify({'error': 'No data available'}), 400

        # Create visualizations
        figures = {}

        # Visitors distribution
        plt.figure(figsize=(10, 6))
        visitor_counts = recommendation_system.df['visitors'].value_counts()
        bars = plt.bar(visitor_counts.index, visitor_counts.values)
        plt.title('Distribution of Visitors', fontsize=16, fontweight='bold')
        plt.xlabel('Visitor Level', fontsize=12)
        plt.ylabel('Count', fontsize=12)

        for bar in bars:
            height = bar.get_height()
            plt.text(bar.get_x() + bar.get_width()/2., height + 0.1,
                    f'{int(height)}', ha='center', va='bottom', fontweight='bold')

        plt.tight_layout()

        # Convert to base64
        buf = io.BytesIO()
        plt.savefig(buf, format='png')
        buf.seek(0)
        figures['visitors_distribution'] = base64.b64encode(buf.getvalue()).decode('utf-8')
        plt.close()

        return jsonify({'visualizations': figures})

    except Exception as e:
        return jsonify({'error': str(e)}), 500

if __name__ == '__main__':
    # Initialize the system when starting the server
    recommendation_system.initialize_system()
    app.run(host='0.0.0.0', port=5002, debug=True)
```

APPENDIX D.

Evaluation Tool

Cosine Similarity Algorithm

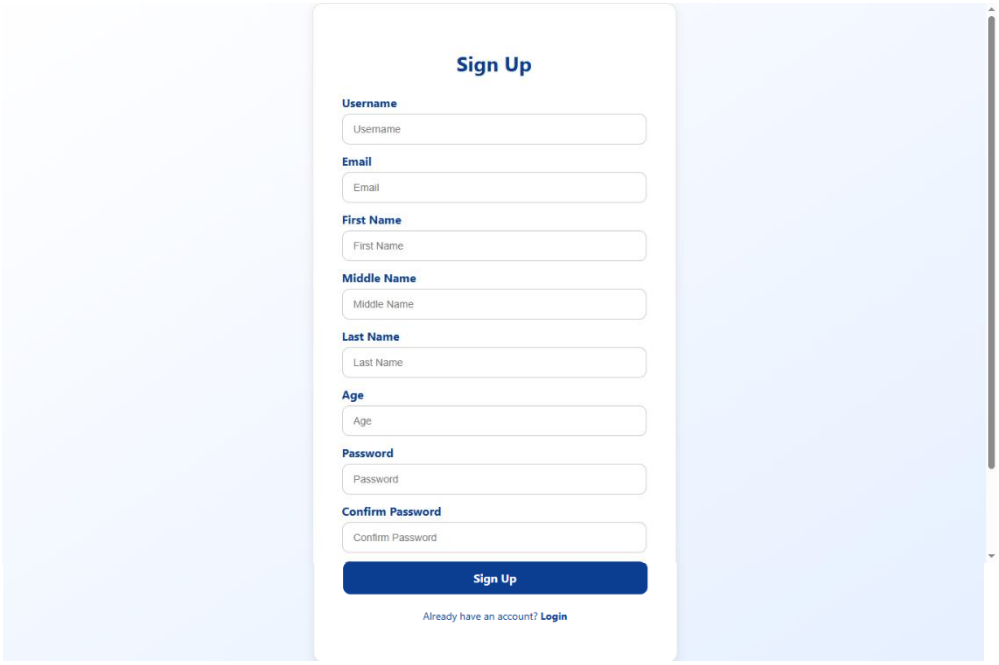
$$similarity(A, B) = cos(\theta) = \frac{A \cdot B}{||A||\,||B||} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sum_{i=1}^{n} B_i^2}.$$

APPENDIX E.

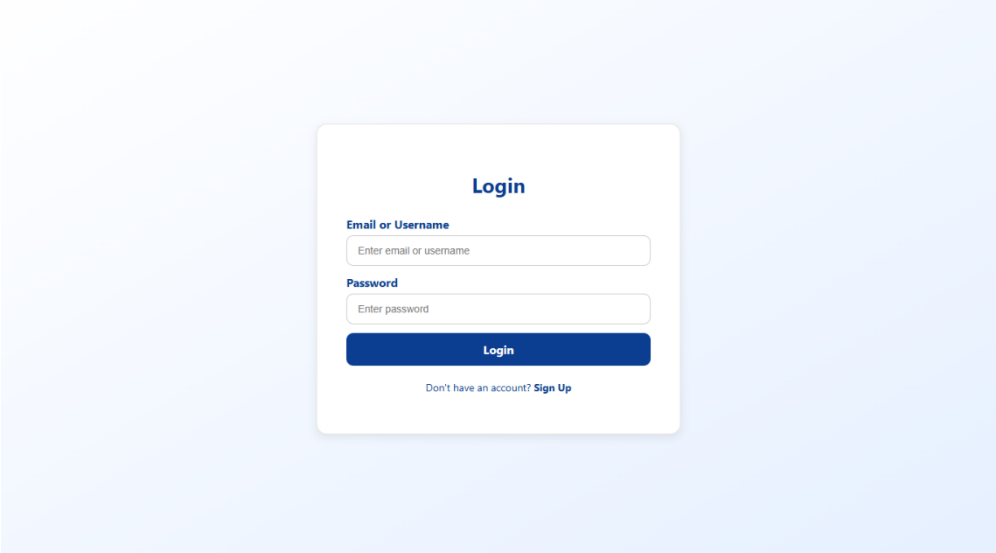
Sample Input / Output

Storyboard for Tourist

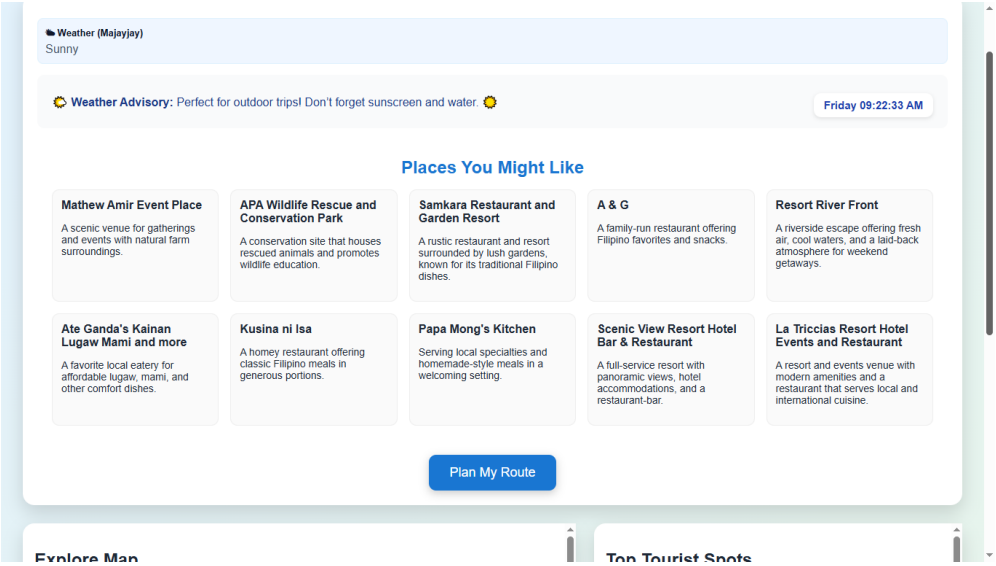
This module shows where the tourist can register their account



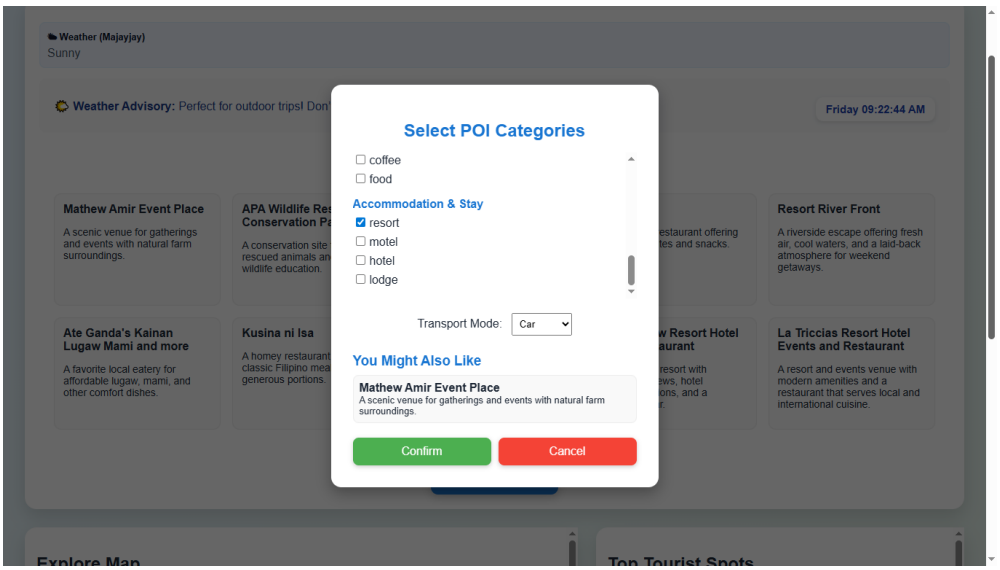
This is where the user can login and access the system



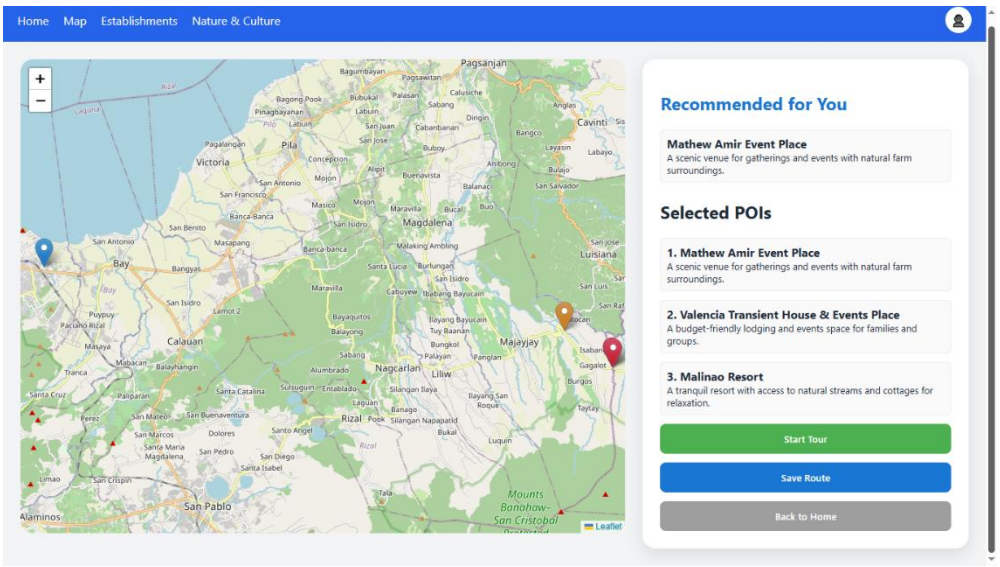
This is the user interface where he can view places and generate a route from the system. Additionally the system would recommend the user based on his feedback on places



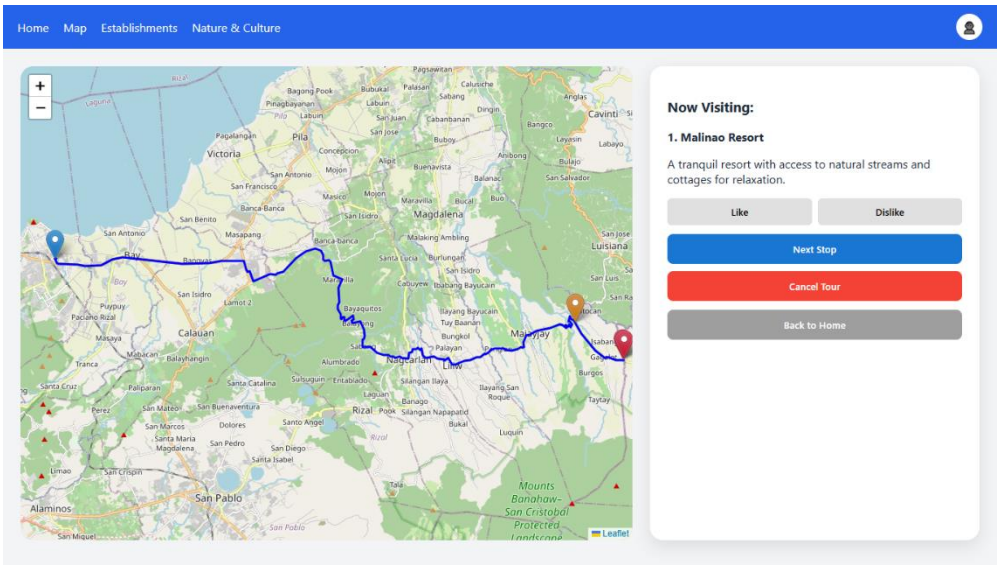
After clicking the plan my route button the user can select what type of places he wants to visit



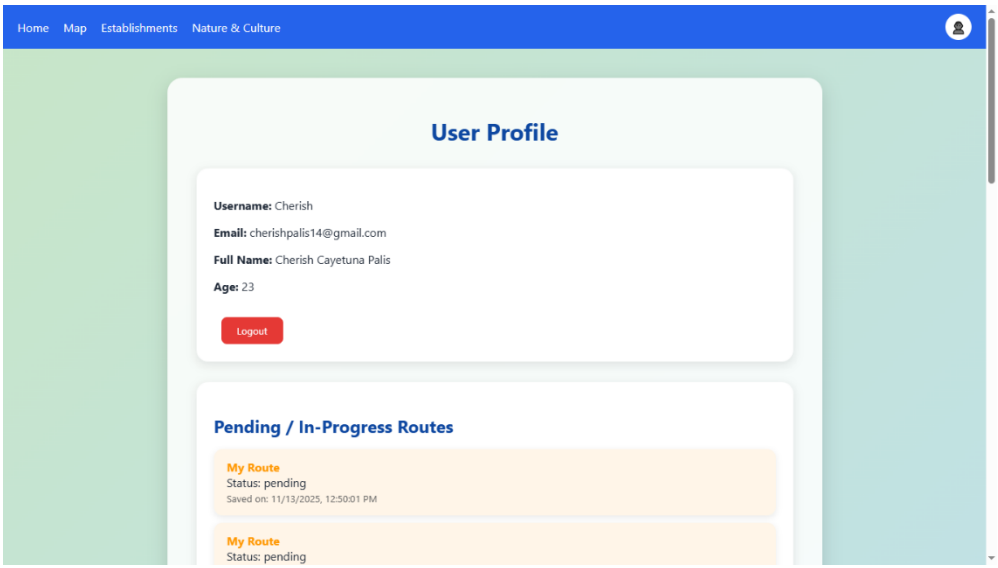
The user would be then show the places with the selected tags and the user can remove or arrange the places he/she plans to visit

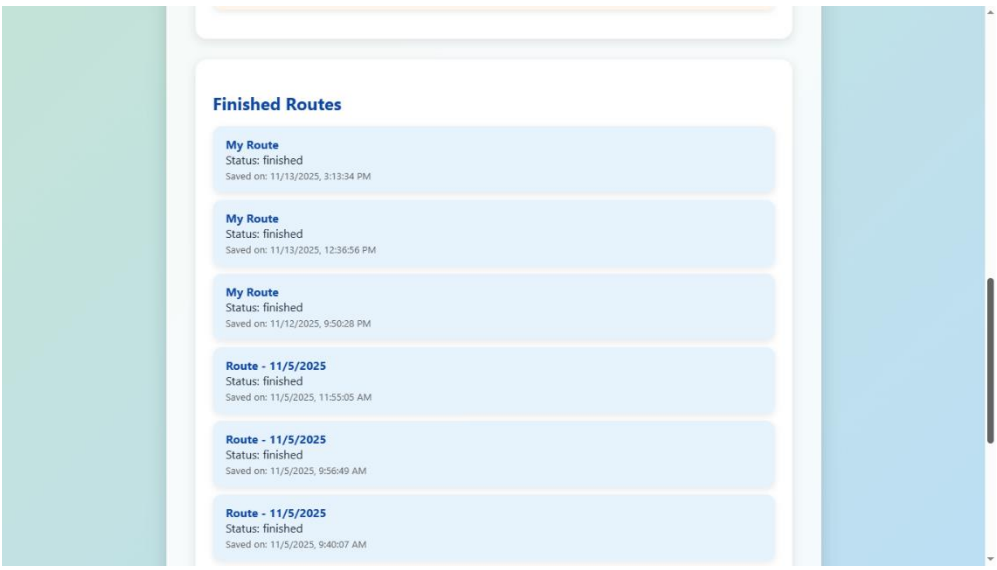


Once the user starts the tour a route (blue line) would appear and the user can follow the line to go to the selected point



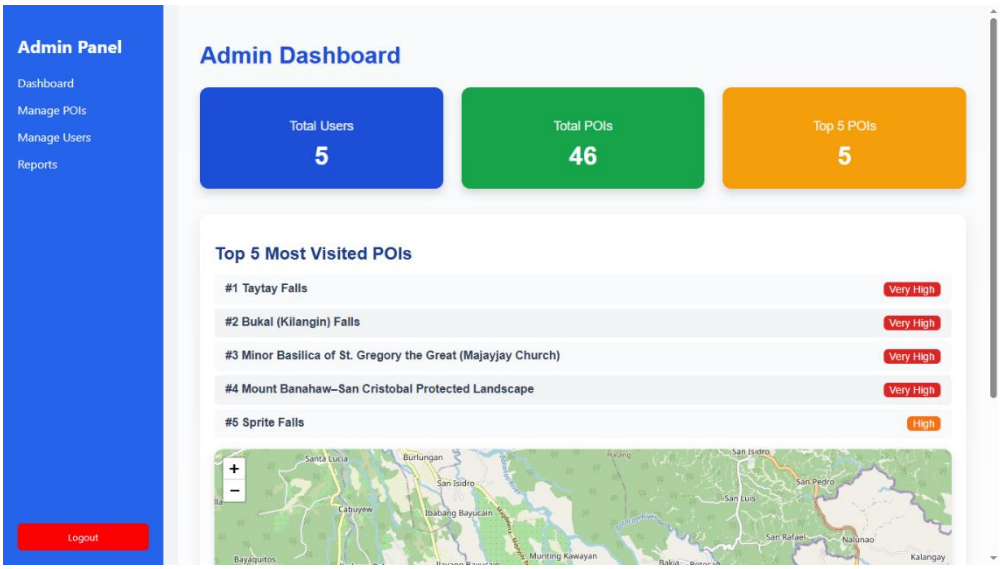
After finishing the tour he can saved the tour and view it on his profile page



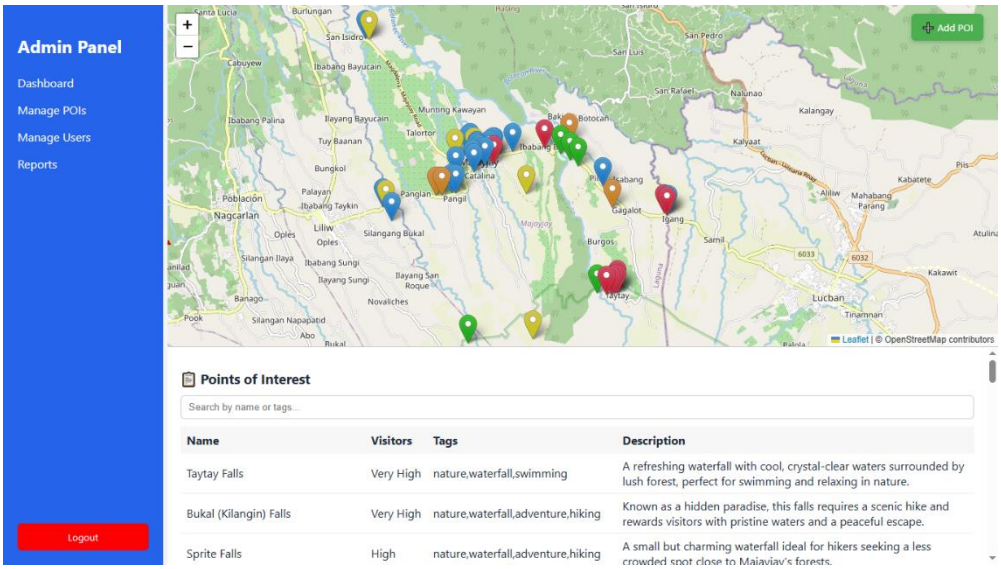


Storyboard for Admin

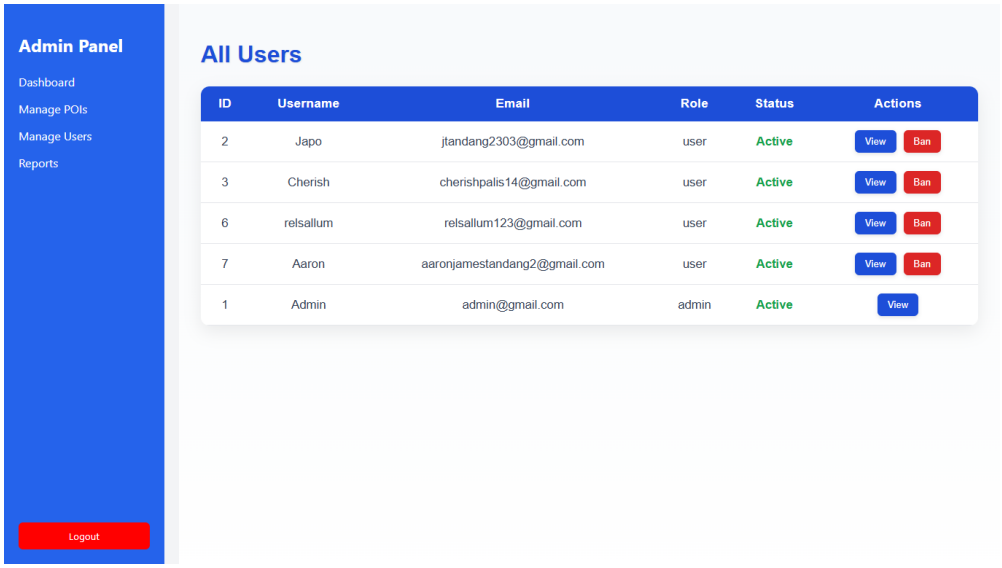
This module shows the admin dashboard where they can see the total users, total places, top pois



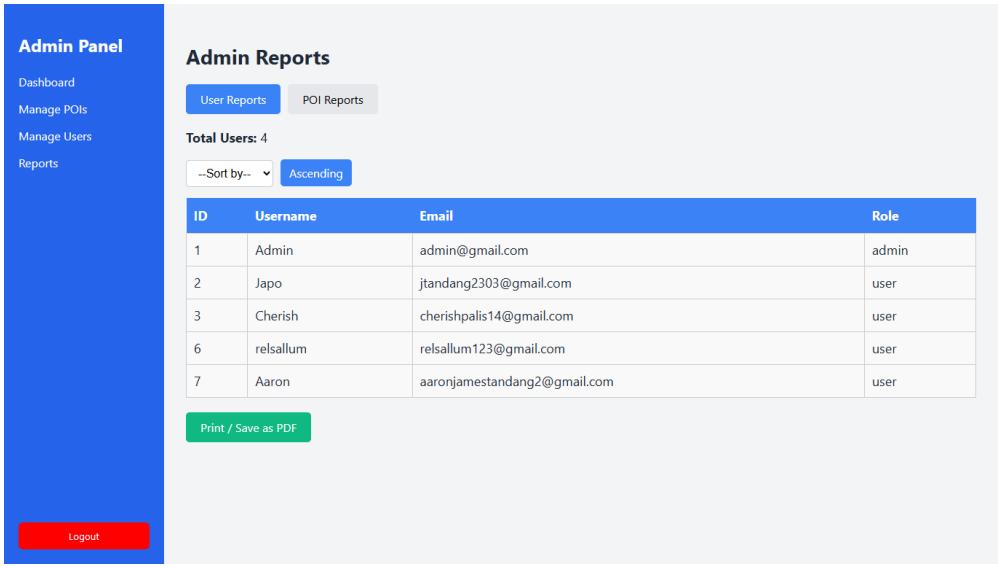
The admin can then add a place, edit an existing place, or remove a place



The admin can either view a user account or ban him



The admin can generate a report on total user and places.



APPENDIX F.

Turnitin Result

Turnitin Result 12%



APPENDIX G.

Publishable Paper (IEEE format)

IMPLEMENTING TOURISM CIRCUIT CONCEPT WITH MACHINE LEARNING ON A WEB-BASED TOURISM PLATFORM FOR MAJAYJAY, LAGUNA

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Article Info

Article history:

Keywords:
Tourism Circuit, Web-Based Platform, Machine Learning, Sustainable Tourism, Majayjay Laguna, Interactive Mapping, Community-Based Tourism

ABSTRACT

Tourism plays a vital role in local economic development, especially in areas with rich cultural and natural attractions. Majayjay, Laguna, known as a hidden gem, continues to face challenges in promoting its tourist spots because of limited online presence and fragmented information. To address this, a web-based tourism platform with machine learning was developed to improve the visibility, accessibility, and usability of the Majayjay tourism circuit. The system features user registration, accommodation listings, booking, and personalized recommendations that guide tourists in planning their visits. It was built using React.js for the frontend, Node.js for backend operations, Flask (Python) for the machine learning component, Leaflet.js for interactive mapping, and MySQL for data management. The machine learning model was trained and deployed locally, ensuring flexibility and independence from third-party platforms. By integrating interactive maps and intelligent recommendations, the platform enhances the tourist experience while promoting sustainable tourism practices. Ultimately, this project demonstrates how digital innovation can strengthen community-based tourism and create opportunities for local economic growth.

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I. INTRODUCTION

Tourism is one of the world's fastest-growing industries and a major contributor to employment and national income. In the Philippines, tourism accounted for 8.6% of the national GDP in 2023 and welcomed over 5.45 million international visitors, highlighting its significant role in economic development, particularly in culturally and naturally rich provinces.

One strategy used to strengthen local tourism is the tourism circuit, which connects multiple destinations into a structured travel route that enhances visitor experience and promotes regional development. Majayjay, Laguna, with its waterfalls, natural springs, and historical landmarks, is a promising location for such a circuit. Despite its rich cultural and natural assets, the town remains under-promoted and is considered a hidden tourism destination.

Sustainable and ecotourism practices emphasize environmental protection, community participation, and long-term economic benefits, aligning with the United Nations Sustainable Development Goal 8: Decent Work

and Economic Growth. However, limited digital visibility remains a challenge for rural destinations like Majayjay.

With the growing reliance on technology in travel planning, integrating Information Technology into tourism development offers a practical solution. A web-based tourism platform enhanced with Geographic Information Systems (GIS) and machine learning can improve destination visibility, provide personalized recommendations, and support local businesses. By combining tourism and digital innovation, Majayjay's tourism circuit can be promoted effectively while ensuring sustainable community development.

II. METHODOLOGY

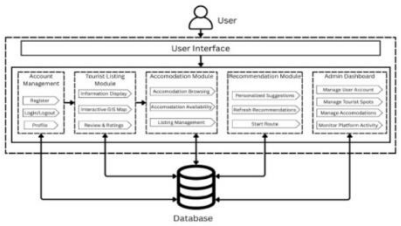


Figure 1. System Architecture

2.1 Project Design

This section introduces the overall design and architecture of the proposed web-based tourism platform for Majayjay, Laguna. It outlines the system structure, major components, and core functionalities. The project follows a modular and layered design approach to ensure scalability, performance, and ease of maintenance. The system integrates machine learning for personalized tourist experiences and employs a client-server model to manage interactions between users, server processes, and the database.

Materials

React.js was the main tool we used for building the system's interface. It allowed us to design pages that are interactive, easy to use, and responsive on different devices. Through React, we were able to organize the platform into reusable parts, which made development faster and more manageable. This helped us create a user-friendly experience for tourists exploring the system.

Node.js was used to connect the different parts of the system and make sure everything worked together properly. It served as the middle layer that handled requests, such as saving bookings, retrieving tourist spot details, and passing information between the website and the database. Its reliability made it suitable for managing the platform's activities smoothly.

Flask (Python) supported the machine learning component of our project. We used it to run our recommendation system, which suggests tourist spots based on user preferences and behaviors. Flask made it easier to connect our machine learning models with the main website so users could receive personalized recommendations in real time.

XAMPP and MySQL were used for storing and managing important information in the system. XAMPP provided a local environment where we could test the website before deployment, while MySQL served as the database where we kept records such as user accounts, tourist attractions, accommodations, and bookings. These tools were free, practical, and reliable for our needs.

Leaflet.js was used to display interactive maps in the platform. With this tool, we were able to show the locations of tourist spots, accommodations, and other points of interest in Majayjay, Laguna. It allowed users to easily explore the area visually and plan their trips more effectively. Since it is open-source and lightweight, it was a cost-free and efficient way to add mapping features to the system.

Project Testing Evaluation

Evaluation Procedure



Figure 9. Product Quality Model

The ISO 25010 model was applied in this study to evaluate the overall quality of the system. ISO 25010 focuses on eight key quality characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Figure 9 illustrates the ISO 25010 quality model, ensuring that the tourism platform is useful, dependable, efficient, and safe forusers.

Population of the Study

The researchers will select 100 respondents and 3 IT experts to participate. Respondents will be guided during testing to ensure the system works properly, while

IT experts will focus on checking data security and system performance.

Type of Respondents	Number of Respondents
User	100
IT Expert	3
TOTAL	103

III. RESULT AND DISCUSSION

This chapter presents the result and discussion to support the methodology utilized in the achievement of the objectives.

3.1 Results by Objectives of the Study

Figure 11 illustrates the main interface of the Majayjay Tourism Circuit Web Platform, showcasing its core features designed to support efficient tourist exploration. The interface provides easy access to Points of Interest (POIs), lodging listings, itinerary planning, and the user dashboard. The layout is organized and intuitive, ensuring accessibility and ease of navigation for both tourists and local users.

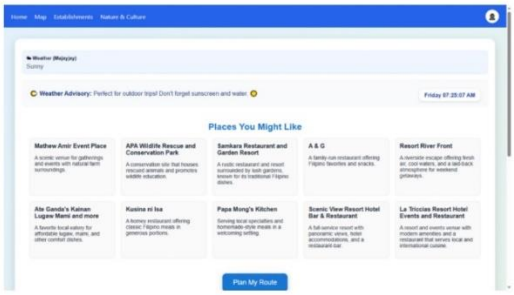
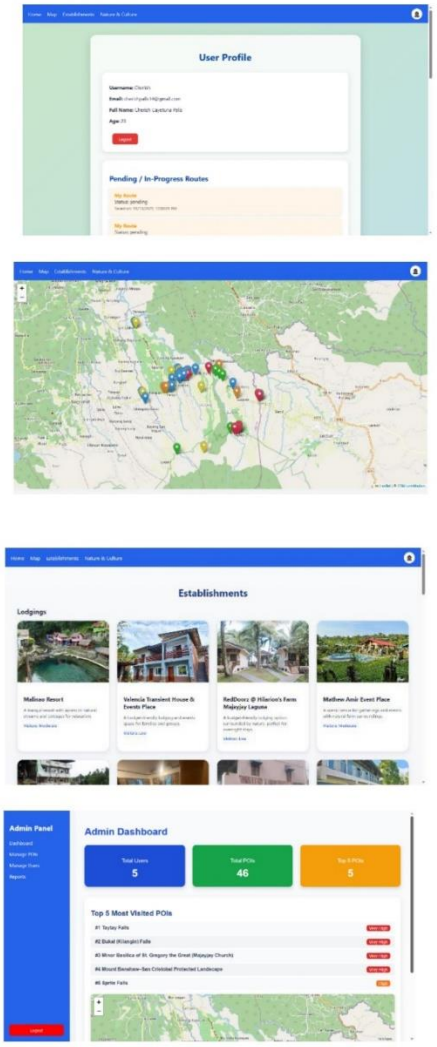


Figure 11. User Interface

Figure 12 presents the key system modules. The User Account Module (a) enables secure registration, login, and profile management, allowing users to save preferences and itineraries through token-based authentication. The Tourism Listings Module (b) displays Majayjay's major attractions with detailed descriptions, images, and map-based visualization. The Lodging Information Module (c) provides accommodation details such as pricing, amenities, and proximity to tourist spots, assisting users in planning their stay. The Administrative Dashboard (d) allows system administrators to manage destinations, lodging data, user accounts, and system reports, ensuring platform reliability and content accuracy.



a

b

c

d

Figure 12.User Interface for recommendation system (a) Account Module, (b) Tourist Listing Module, (c) Lodging Information, and (d) Admin Dashboard

Figure 13 shows the recommendation interface, highlighting the platform's machine learning–based capability to generate personalized tourist recommendations. Suggested destinations are displayed based on user preferences, saved itineraries, and previous interactions. Figure 14 further demonstrates the system's adaptability by presenting recommendations tailored to

rainy weather conditions, indicating the integration of contextual and seasonal factors into the recommendation process.

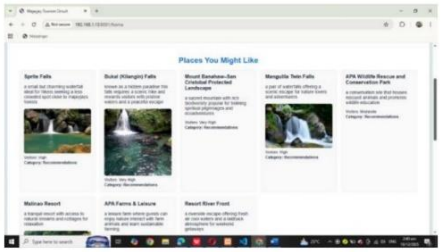


Figure 13. Recommendation Interface

Figure 14 presents the recommendation results under rainy weather conditions. This highlights how the machine learning system incorporates seasonal factors into its decision-making process, enabling it to generate more accurate, context-aware, and user-friendly recommendations

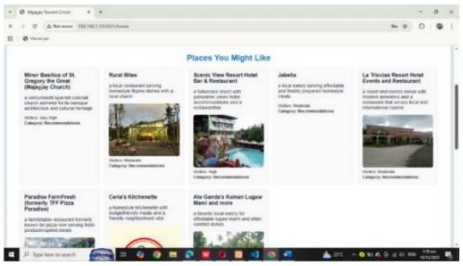


Figure 14. Smart Recommendation based on Season

System testing and evaluation were conducted using ISO 25010 and the Technology Acceptance Model (TAM). ISO 25010 assessed the system's functionality, usability, reliability, efficiency, maintainability, and security, while TAM evaluated user acceptance in terms of perceived usefulness and ease of use. The evaluation results indicate that the platform meets quality standards and is positively accepted by users.

Table 1. Technology Acceptance Model (TAM) Results

Characteristics	Mean	Standard Deviation	interpretation
Perceived usefulness	4.28	0.85	Strongly Agree
Perceived Ease of Use	4.21	0.92	Strongly Agree
Attitude	4.30	0.86	Strongly Agree
Behavioural Intention	4.18	0.85	Agree

Table I shows the system evaluation based on TAM characteristics. Perceived usefulness (4.28), perceived ease of use (4.21), and attitude (4.30) were all "Strongly Agree," indicating that users found the system useful, easy to use, and favorable. Behavioral intention scored 4.18 ("Agree"), showing a slightly lower but still positive intention to use the system. Standard deviations (0.85–0.92) suggest consistent responses among participants. Overall, the results indicate that the system meets key quality and usability expectations.

Table 2. ISO 25010 Result

Characteristics	Mean	Standard Deviation	interpretation
Functional Sustainability	4.59	0.22	Very Satisfied
Performance Efficiency	4.58	0.17	Very Satisfied
Compatibility	4.50	0.28	Very Satisfied
Usability	4.75	0.18	Very Satisfied
Reliability	4.75	0.17	Very Satisfied
Security	4.67	0.22	Very Satisfied
Maintainability	4.67	0.22	Very Satisfied
Portability	5.00	0	Very Satisfied

Table 2.shows the evaluation of the tourism platform based on ISO 25010 characteristics. All characteristics scored between 4.50 and 5.00, indicating that users were very satisfied with the system. High scores in Usability (4.75) and Reliability (4.75) suggest that the platform is user-friendly and operates stably without errors. Portability scored perfectly (5.00), showing that the system can be deployed across multiple environments seamlessly.

VI. CONCLUSION AND FUTURE WORKS

The Majayjay Tourism Circuit platform successfully provides tourists with easy access to information on attractions, accommodations, and activities, highlighting the town as a hidden gem. Developed using Agile Scrum, the platform ensures iterative improvements and user-centered design. Its five modules—Account, Tourist Listing, Accommodation Management, Recommendation, and Admin Management—were tested and confirmed to function reliably, efficiently, and compatibly across devices and browsers.

TAM evaluation showed high user satisfaction in usability, usefulness, security, and portability, with users appreciating the interactive map and personalized recommendations. Overall, the platform is reliable, user-friendly, and effective in promoting tourism, offering a valuable tool for both tourists and administrators.

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Recommendation

Based on the findings from system testing, compatibility testing, and the Technology Acceptance Model (TAM) evaluation, several recommendations can be made to further improve the Majayjay Tourism Circuit platform.

First, the Recommendation Module can be enhanced to provide more personalized suggestions based on user behavior, preferences, and past activities. Adding features such as ratings or reviews from other tourists can also improve the accuracy and usefulness of the recommendations.

Second, the platform's compatibility can be strengthened by conducting formal testing across a wider range of devices, screen sizes, and browsers. This will ensure that all users, regardless of their device, have a seamless experience. Special attention should be given to mobile platforms and newer browsers, as these are commonly used by tourists today.

Third, improvements in user interface and usability can make the platform more engaging. Interactive elements like virtual tours, image galleries, and detailed route maps can enhance the user experience, while tooltips or short tutorials can help new users navigate all the features more easily.

Fourth, regular system maintenance and updates are recommended to ensure the platform remains reliable, secure, and up-to-date. This includes monitoring system performance, addressing any bugs, and updating tourist and accommodation information based on user feedback.

Finally, the platform can be used more effectively to promote Majayjay tourism by including promotional content, seasonal events, and special packages. Collaboration with local businesses to offer real-time deals for accommodations, restaurants, and tours can further attract visitors. Maintaining high standards for data security and privacy, including protecting user login credentials and considering additional security measures, will also strengthen user trust and confidence in the platform.

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APPENDIX H.

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- **Getting Started with Generative AI**
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- **Make Agentic AI Work for You**
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